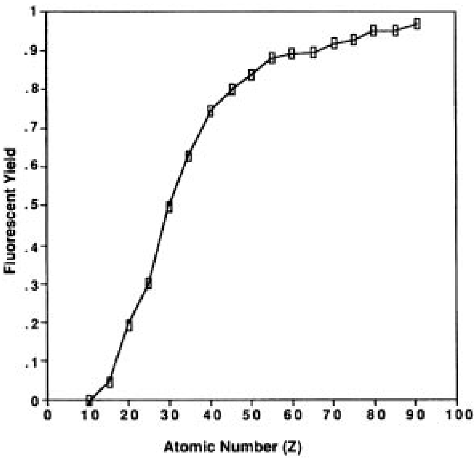
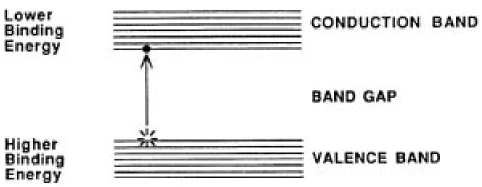


MARGIN FIGURE 2-6
Fluorescence yield for the K shell versus atomic number.



MARGIN FIGURE 2-7
Energy level diagram for solids. An electron promoted from the valence band to the conduction band may move freely in the material to constitute an electric current.

Superconductivity was first discovered in mercury, by Onnes in 1911.[24] However, a satisfactory theoretical explanation of superconductivity, using the formalism of quantum mechanics, did not evolve until 1957 when BCS theory was proposed by Bardeen, Cooper, and Schrieffer.[25, 26]

The 1987 Nobel Prize in Physics was awarded jointly to J. George Bednorz and K. Alexander Müller "for their important breakthrough in the discovery of superconductivity in ceramic materials."

band gap of the solid. Semiconductors have many applications in radiation detection and are discussed further in Chapter 8.

SUPERCONDUCTIVITY

In a conductor, very little energy is required to promote electrons to the conduction band. However, free movement of electrons in the conduction band is not guaranteed. As electrons attempt to move through a solid, they interact with other electrons and with imperfections such as impurities in the solid. At each "encounter" some energy is transferred to atoms and molecules and ultimately appears as heat. This transfer of energy (the basis of electrical "resistance") is usually viewed as energy "loss." Sometimes, however, this "loss" is the primary goal of electrical transmission as, for example, in an electric iron or the filament of an x-ray tube.

There are materials in which, under certain conditions, there is no resistance to the flow of electrons. These materials are called superconductors. In a superconductor the passage of an electron disturbs the structure of the material in such a way as to encourage the passage of another electron arriving after exactly the right interval of time. The passage of the first electron establishes an oscillation in the positive charge of the material that "pulls" the second electron through. This behavior has been compared to "electronic water skiing" where one electron is swept along in another electron's "wake." Thus electrons tend to travel in what are known as "Cooper pairs."[26] Cooper pairs do not (in a probabilistic sense) travel close to each other. In fact, many other electrons can separate a Cooper pair. Cooper pair electrons are also paired with other electrons. It has been shown mathematically that the only possible motion of a set of interleaved pairs of electrons is movement as a unit. In this fashion, the electrons do not collide randomly with each other and "waste" energy. Instead, they travel as a unit with essentially no resistance to flow. Currents have been started in superconducting loops of wire that have continued for several years with no additional input of energy.

Many types of materials exhibit superconductive behavior when cooled to temperatures in the range of a few degrees Kelvin (room temperature is approximately 295°K). Lowering the temperature of some solids promotes superconductivity by decreasing molecular motion, thereby decreasing the kinetic energy of the material. Twenty-six elements and thousands of alloys and compounds exhibit this behavior. Maintenance of materials at very low temperatures requires liquid helium as a cooling agent. Helium liquefies at 23°K, is relatively expensive, and is usually insulated from ambient conditions with another refrigerant such as liquid nitrogen.

On theoretical grounds it had been suspected for many years that superconductivity can exist in some materials at substantially higher temperatures, perhaps even room temperature. In January 1986, Bednorz and Müller discovered a ceramic, an oxide of barium, lanthanum, and copper, that is superconducting at temperatures up to 35°K.[27] With this discovery, superconductivity was achieved for the first time at temperatures above liquid helium. This finding opened new possibilities for cheaper and more convenient refrigeration methods. Subsequently, several other ceramics have exhibited superconductivity at temperatures of up to 135°K. There is, at present, no satisfactory theoretical explanation for superconductivity in ceramics. Some believe that there is a magnetic phenomenon that is analogous to the electronic phenomenon cited in BCS theory. The current record for highest superconducting temperature in a metal is in magnesium diboride at 39°K.[28] While this temperature is not as high as has been achieved in some ceramics, this material would be easier to fashion into a wire for use as a winding in a superconducting magnet. These materials have great potential for yielding "perfect" conductors (i.e., with no electrical resistance) that are suitable for everyday use. Realization of this potential would have a profound impact upon the design of devices such as electrical circuits and motors. It would revolutionize fields as diverse as computer science, transportation, and medicine, including radiology.

THE NUCLEUS

Nuclear Energy Levels

A nucleus consists of two types of particles, referred to collectively as nucleons. The positive charge and roughly half the mass of the nucleus are contributed by protons. Each proton possesses a positive charge of $+1.6 \times 10^{-19}$ coulombs, equal in magnitude but opposite in sign to the charge of an electron. The number of protons (or positive charges) in the nucleus is the atomic number of the atom. The mass of a proton is 1.6734×10^{-27} kg. Neutrons, the second type of nucleon, are uncharged particles with a mass of 1.6747×10^{-27} kg. Outside the nucleus, neutrons are unstable and divide into protons, electrons, and antineutrinos (see Chapter 3). The half-life of this transition is 12.8 minutes. Neutrons are usually stable inside nuclei. The number of neutrons in a nucleus is the neutron number N for the nucleus. The mass number A of the nucleus is the number of nucleons (neutrons and protons) in the nucleus. The mass number $A = Z + N$.

The standard form used to denote the composition of a specific nucleus is

$$^{A}_{Z}\mathrm{X}$$

where X is the chemical symbol (e.g., H, He, Li) and A and Z are as defined above. There is some redundancy in this symbolism. The atomic number, Z, is uniquely associated with the chemical symbol, X. For example, when $Z = 6$, the chemical symbol is always C, for the element carbon.

Expressing the mass of atomic particles in kilograms is unwieldy because it would be a very small number requiring scientific notation. The atomic mass unit (amu) is a more convenient unit for the mass of atomic particles. 1 amu is defined as 1/12 the mass of the carbon atom, which has six protons, six neutrons, and six electrons. Also,

$$1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg}$$

The shell model of the nucleus was introduced to explain the existence of discrete nuclear energy states. In this model, nucleons are arranged in shells similar to those available to electrons in the extranuclear structure of an atom. Nuclei are extraordinarily stable if they contain 2, 8, 14, 20, 28, 50, 82, or 126 protons or similar numbers of neutrons. These numbers are termed magic numbers and may reflect full occupancy of nuclear shells. Nuclei with odd numbers of neutrons or protons tend to be less stable than nuclei with even numbers of neutrons and protons. The pairing of similar nucleons increases the stability of the nucleus. Data tabulated below support this hypothesis.

Number of Protons	Number of Neutrons	Number of Stable Nuclei
Even	Even	165
Even	Odd	57
Odd	Even	53
Odd	Odd	6

Nuclear Forces and Stability

Protons have “like” charges (each has the same positive charge) and repel each other by the electrostatic force of repulsion. One may then ask the question, How does a nucleus stay together? The answer is that when protons are very close together, an attractive force comes into play. This force, called the “strong nuclear force,” is 100 times greater than the electrostatic force of repulsion. However, it acts only over distances of the order of magnitude of the diameter of the nucleus. Therefore, protons can stay together in the nucleus once they are there. Assembling a nucleus by forcing protons together requires the expenditure of energy to overcome the electrostatic repulsion. Neutrons, having no electrostatic charge, do not experience the electrostatic force. Therefore, adding neutrons to a nucleus requires much less energy. Neutrons are, however,

Superconductors have zero resistance to the flow of electricity. They also exhibit another interesting property called the Meisner Effect: If a superconductor is placed in a magnetic field, the magnetic field lines flow around it. That is, the superconductor excludes the magnetic field. This can be used to create a form of magnetic levitation.

As of this writing, magnetic resonance imagers are the only devices using the principles of superconductivity that a typical layperson might encounter. Other applications of superconductivity are found chiefly in research laboratories.

The word “atom” comes from the Greek “atomos” which means “uncuttable.”

The half-life for decay of the neutron is 12.8 minutes. This means that in a collection of a large number of neutrons, half would be expected to undergo the transition in 12.8 minutes. Every 12.8 minutes, half the remaining number would be expected to decay. After 7 half-lives (3.7 days), fewer than 1% would be expected to remain as neutrons.

Masses of atomic particles are as follows:

$$\text{Electron} = 0.00055 \text{ amu}$$

$$\text{Proton} = 1.00727 \text{ amu}$$

$$\text{Neutron} = 1.00866 \text{ amu}$$

Note that the proton and neutron have a mass of approximately 1 amu and that the neutron is slightly heavier than the proton.

Quantum Electrodynamics (QED)
Modern quantum mechanics explains a force in terms of the exchange of "messenger particles." These particles pass between (are emitted and then absorbed by) the particles that are affected by the force.
The messenger particles are as follows:

Force	Messenger
Strong nuclear	Gluon
Electromagnetic	Photon
Weak nuclear	"W" and "Z"
Gravity	Graviton

affected by a different "weak nuclear force." The weak nuclear force causes neutrons to change spontaneously into protons plus almost massless virtually noninteracting particles called neutrinos. The opposite transition, protons turning into neutrons plus neutrinos, also occurs. These processes, called beta decay, are described in greater detail in Chapter 3. The fourth of the traditional four "fundamental forces" of nature, gravity, is extremely overshadowed by the other forces within an atom, and thus it plays essentially no role in nuclear stability or instability. The relative strengths of the "four forces" are as follows:

Type of Force	Relative Strength
Nuclear	1
Electrostatic	10^{-2}
Weak	10^{-13}
Gravitational	10^{-39}

Of these forces, the first three have been shown to be manifestations of the same underlying force in a series of "Unification Theories" over recent decades. The addition of the fourth, gravity, would yield what physicists term a "Grand Unified Theory," or GUT.

Nuclear Binding Energy

The mass of an atom is less than the sum of the masses of its neutrons, protons, and electrons. This seeming paradox exists because the binding energy of the nucleus is so significant compared with the masses of the constituent particles of an atom, as expressed through the equivalence of mass and energy described by Einstein's famous equation.[29]

$$E = mc^2$$

The mass difference between the sum of the masses of the atomic constituents and the mass of the assembled atom is termed the *mass defect*. When the nucleons are separate, they have their own individual masses. When they are combined in a nucleus, some of their mass is converted into energy. In Einstein's equation, an energy E is equivalent to mass m multiplied by the speed of light in a vacuum, c (2.998×10^8 m/sec) squared. Because of the large "proportionality constant" c^2 in this equation, one kilogram of mass is equal to a large amount of energy, 9×10^{16} joules, roughly equivalent to the energy released during detonation of 30 megatons of TNT. The energy equivalent of 1 amu is

$$\frac{(1\text{ amu})(1.660 \times 10^{-27}\text{ kg/amu})(2.998 \times 10^{8}\text{ m/sec})^2}{(1.602 \times 10^{-13}\text{ J/Mev})} = 931\text{ MeV}$$

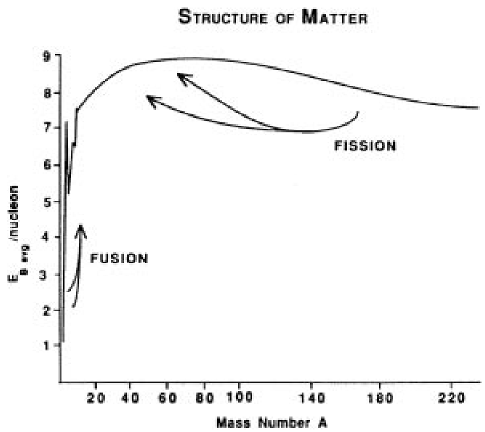
The binding energy (mass defect) of the carbon atom with six protons and six neutrons (denoted as ${}^{12}_{6}\mathrm{C}$) is calculated in Example 2-2.

Example 2-2

$$\begin{aligned}
\text{Mass of 6 protons} &= 6(1.00727\text{ amu}) = 6.04362\text{ amu}\\
\text{Mass of 6 neutrons} &= 6(1.00866\text{ amu}) = 6.05196\text{ amu}\\
\text{Mass of 6 electrons} &= 6(0.00055\text{ amu}) = 0.00330\text{ amu}\\
\text{Mass of components of } {}^{12}_{6}\mathrm{C} &= 12.09888\text{ amu}\\
\text{Mass of } {}^{12}_{6}\mathrm{C}\text{ atom} &= 12.00000\text{ amu}\\
\text{Mass defect} &= 0.09888\text{ amu}\\
\text{Binding energy of } {}^{12}_{6}\mathrm{C}\text{ atom} &= (0.09888\text{ amu})(931\text{ MeV/amu})\\
&= 92.0\text{ MeV}
\end{aligned}$$

Almost all of this binding energy is associated with the $^{12}_{6}C$ nucleus. The average binding energy per nucleon of $^{12}_{6}C$ is 92.0 MeV per 12 nucleons, or 7.67 MeV per nucleon.

In Margin Figure 2-8 the average binding energy per nucleon is plotted as a function of the mass number A.



MARGIN FIGURE 2-8
Average binding energy per nucleon versus mass number.

NUCLEAR FISSION AND FUSION

Energy is released if a nucleus with a high mass number separates or fissions into two parts, each with an average binding energy per nucleon greater than that of the original nucleus. The energy release occurs because such a split produces low-Z products with a higher average binding energy per nucleon than the original high-Z nucleus (Margin Figure 2-8). A transition from a state of lower "binding energy per nucleon" to a state of higher "binding energy per nucleon" results in the release of energy. This is reminiscent of the previous discussion of energy release that accompanies an L to K electron transition. However, the energy available from a transition between nuclear energy levels is orders of magnitude greater than the energy released during electron transitions.

Certain high-A nuclei (e.g., ^{235}U, ^{239}Pu, ^{233}U,) fission spontaneously after absorbing a slowly moving neutron. For ^{235}U, a typical fission reaction is

$$^{235}_{92}\mathrm{U} + \text{neutron} \rightarrow {}^{236}_{92}\mathrm{U} \rightarrow {}^{92}_{36}\mathrm{Kr} + {}^{141}_{56}\mathrm{Ba} + 3 \text{ neutrons} + Q$$

The energy released is designated as Q and averages more than 200 MeV per fission. The energy is liberated primarily as γ radiation, kinetic energy of fission products and neutrons, and heat and light. Products such as $^{92}_{36}Kr$ and $^{141}_{56}Ba$ are termed fission by-products and are radioactive. Many different by-products are produced during fission. Neutrons released during fission may interact with other ^{235}U nuclei and create the possibility of a chain reaction, provided that sufficient mass of fissionable material (a critical mass) is contained within a small volume. The rate at which a material fissions may be regulated by controlling the number of neutrons available each instant to interact with fissionable nuclei. Fission reactions within a nuclear reactor are controlled in this way. Uncontrolled nuclear fission results in an "atomic explosion."

"The energy produced by the breaking down of the atom is a very poor kind of thing. Anyone who expects a source of power from the transformation of these atoms is talking moonshine."
E. Rutherford, 1933.

The critical mass of ^{235}U is as little as 820 g if in aqueous solution or as much as 48.6 kg if a bare metallic sphere.[30]

Nuclear fission was observed first in 1934 during an experiment conducted by Enrico Fermi.[31, 32] However, the process was not described correctly until publication in 1939 of analyses by Hahn and Strassmann[33] and Meitner and Frisch.[34] The first controlled chain reaction was achieved in 1942 at the University of Chicago. The first atomic bomb was exploded in 1945 at Alamogordo, New Mexico.[35]

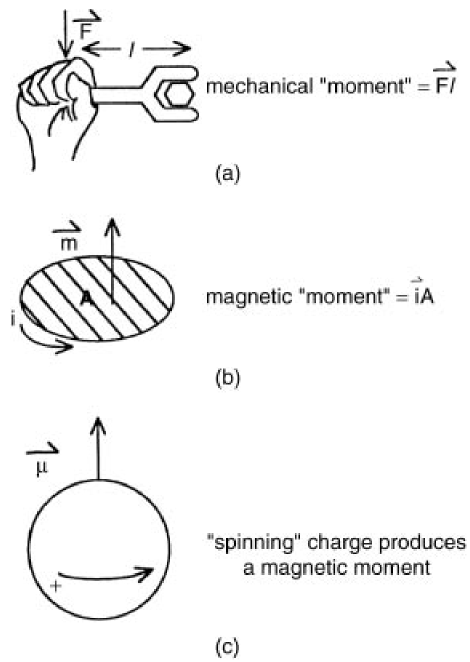
Certain low-mass nuclei may be combined to produce a nucleus with an average binding energy per nucleon greater than that for either of the original nuclei. This process is termed nuclear fusion (Margin Figure 2-8) and is accompanied by the release of large amounts of energy. A typical reaction is

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + \text{neutron} + Q$$

In this particular reaction, $Q = 18$ MeV.

To form products with higher average binding energy per nucleon, nuclei must be brought sufficiently near one another that the nuclear force can initiate fusion. In the process, the strong electrostatic force of repulsion must be overcome as the two nuclei approach each other. Nuclei moving at very high velocities possess enough momentum to overcome this repulsive force. Adequate velocities may be attained by heating a sample containing low-Z nuclei to a temperature greater than 12×10^6 °K, roughly equivalent to the temperature in the inner region of the sun. Temperatures this high may be attained in the center of a fission explosion. Consequently, a fusion (hydrogen) bomb is "triggered" with a fission bomb. Controlled nuclear fusion has not

The only nuclear weapons used in warfare were dropped on Japan in 1945. The Hiroshima bomb used fissionable uranium, and the Nagasaki bomb used plutonium. Both destroyed most of the city on which they fell, killing more than 100,000 people. They each released energy equivalent to about 20,000 tons of TNT. Large fusion weapons (H-bombs) release up to 1000-fold more energy.



MARGIN FIGURE 2-9
The concept of a "moment" in physics. **A**: A mechanical moment is defined by force **F** and length l. **B**: A magnetic moment is defined by current i and the area A enclosed by the current. **C**: The magnetic moment produced by a spinning charged object.

The magnetic moment of the proton was first observed by Stern and colleagues in 1933.[36, 37] The magnetic moment of the neutron was measured by Alvarez and Bloch in 1940.[38] Bloch went on to write the fundamental equations for the "relaxation" of nuclear spins in a material in a static magnetic field that has been perturbed by radiofrequency energy. These equations are the basis of magnetic resonance imaging.

yet been achieved on a macroscopic scale, although much effort has been expended in the attempt.

NUCLEAR SPIN AND NUCLEAR MAGNETIC MOMENTS

Protons and neutrons behave like tiny magnets and are said to have an associated magnetic moment. The term *moment* has a strict meaning in physics. When a force is applied on a wrench to turn a bolt (Margin Figure 2-9A), for example, the mechanical moment is the product of force and length. The mechanical moment can be increased by increasing the length of the wrench, applying more force to the wrench, or a combination of the two. A magnetic moment (Margin Figure 2-9B) is the product of the current in a circuit (a path followed by electrical charges) and the area encompassed by the circuit. The magnetic moment is increased by increasing the current, the area, or a combination of the two. The magnetic moment is a vector, a quantity having both magnitude and direction.

Like electrons, protons have a characteristic called "spin," which can be explained by treating the proton as a small object spinning on its axis. In this model, the spinning charge of the proton produces a magnetic moment (Margin Figure 2-9C).

The "spinning charge" model of the proton has some limitations. First, the mathematical prediction for the value of the magnetic moment is not equal to what has been measured experimentally. From the model, a proton would have a fundamental magnetic moment known as the nuclear magneton, u_n:

$$u_n = \frac{e\hbar}{2m_p} = 3.1525 \times 10^{-12} \text{ eV/gauss}$$

where e is the charge of the proton in coulombs, $\hbar$ is Planck's constant divided by 2π, and m_p is the mass of the proton. The magnetic moment magneton, u_p of the proton, however, is

$$u_p = \text{magnetic moment of the proton} = 2.79u_n$$

The unit of the nuclear magneton, energy (electron volt) per unit magnetic field strength (gauss), expresses the fact that a magnetic moment has a certain (potential) energy in a magnetic field. This observation will be used later to describe the fundamental concepts of magnetic resonance imaging (MRI).

The second difficulty of the spinning proton model is that the neutron, an uncharged particle, also has a magnetic moment. The magnetic moment of the neutron equals $1.91u_n$. The explanation for the "anomalous" magnetic moment of the neutron, as well as the unexplained value of the proton's magnetic moment, is that neutrons and protons are not "fundamental" particles. Instead, they are each composed of three particles called quarks[39] that have fractional charges that add up to a unit charge. Quarks do not exist on their own but are always bound into neutrons, protons, or other particles. The presence of a nonuniform distribution of spinning charges attributable to quarks within the neutron and proton explains the observed magnetic moments.

When magnetic moments exist near each other, as in the nucleus, they tend to form pairs with the vectors of the moments pointed in opposite directions. In nuclei with even numbers of neutrons and protons (i.e., even Z, even N), this pairing cancels out the magnetic properties of the nucleus as a whole. Thus an atom such as ${}^{12}_{6}C$ with 6 protons and 6 neutrons has no net magnetic moment because the neutrons and protons are "paired up."

An atom with an odd number of either neutrons or protons will have a net magnetic moment. For example, ${}^{13}_{6}C$ with 6 protons and 7 neutrons has a net magnetic moment because it contains an unpaired neutron. Also, ${}^{14}_{7}N$ with 7 protons and 7 neutrons has a small net magnetic moment because both proton and neutron numbers are odd and the "leftover" neutron and proton do not exactly cancel each other's moments. Table 2-3 lists a number of nuclides with net magnetic moments. The

TABLE 2-3 Nuclides with a Net Magnetic Moment[a]

Nuclide	*Number of Protons*	*Number of Neutrons*	*Magnetic Moment (Multiple of u_n)*
^{1}H	1	0	2.79
^{2}H	1	1	0.86
^{13}C	6	7	0.70
^{14}N	7	7	0.40
^{17}O	8	9	−1.89
^{19}F	9	10	2.63
^{23}Na	11	12	2.22
^{31}P	15	16	1.13
^{39}K	19	20	0.39

[a] Data from Heath, R. L. Table of the Isotopes, in Weast, R. C., et al.(eds.), *Handbook of Chemistry and Physics*, 52nd edition. Cleveland, Chemical Rubber Co., 1972, pp. 245–256.

"Virtual particles" such as quark–antiquark pairs, or "messenger particles" that carry forces between particles, can appear and disappear during a time interval Δt so long as $\Delta t < h/\Delta E$. ΔE is the change in mass/energy of the system that accounts for the appearance of the particles, and h is Planck's constant. This formula is one version of Heisenberg's Uncertainty Principle. In the nucleus, the time interval is the order of magnitude of the time required for a beam of light to cross the diameter of a single proton.

presence of a net magnetic moment for the nucleus is essential to magnetic resonance imaging (MRI). Only nuclides with net magnetic moments are able to interact with the intense magnetic field of a MRI unit to provide a signal to form an image of the body.

NUCLEAR NOMENCLATURE

Isotopes of a particular element are atoms that possess the same number of protons but a varying number of neutrons. For example, ^{1}H (protium), ^{2}H (deuterium), and ^{3}H (tritium) are isotopes of the element hydrogen, and $^{9}_{6}C$, $^{10}_{6}C$, $^{11}_{6}C$, $^{12}_{6}C$, $^{13}_{6}C$, $^{14}_{6}C$, $^{15}_{6}C$, and $^{16}_{6}C$ are isotopes of carbon. An isotope is specified by its chemical symbol together with its mass number as a left superscript. The atomic number is sometimes added as a left subscript.

Isotones are atoms that possess the same number of neutrons but a different number of protons. For example, $^{5}_{2}He$, $^{6}_{3}Li$, $^{7}_{4}Be$, $^{8}_{5}B$, and $^{9}_{6}C$ are isotones because each isotope contains three neutrons. *Isobars* are atoms with the same number of nucleons but a different number of protons and a different number of neutrons. For example, $^{6}_{2}He$, $^{6}_{3}Li$, and $^{6}_{4}Be$ are isobars ($A = 6$). *Isomers* represent different energy states for nuclei with the same number of neutrons and protons. Differences between isotopes, isotones, isobars, and isomers are illustrated below:

The general term for any atomic nucleus, regardless of A, Z, or N, is "nuclide."

	Atomic No. Z	**Neutron No. N**	**Mass No. A**
Isotopes	Same	Different	Different
Isotones	Different	Same	Different
Isobars	Different	Different	Same
Isomers	Same	Same	Same (different nuclear energy states)

The full explanation of nuclear spin requires a description of the three main types of quark (up, down, and strange) along with the messenger particles that carry the strong nuclear force between them (gluons) together with the short-lived "virtual" quarks and antiquarks that pop in and out of existence over very short time periods within the nucleus.[39]

PROBLEMS

*2-1. What are the atomic and mass numbers of the oxygen isotope with 16 nucleons? Calculate the mass defect, binding energy, and binding energy per nucleon for this nuclide, with the assumption that the entire mass defect is associated with the nucleus. The mass of the atom is 15.9949 amu.

*2-2. Natural oxygen contains three isotopes with atomic masses of 15.9949, 16.9991, and 17.9992 and relative abundances of 2500:1:5, respectively. Determine to three decimal places the average atomic mass of oxygen.

*For problems marked with an asterisk, answers are provided on page 491.

2-3. Using Table 2-1 as an example, write the quantum numbers for electrons in boron ($Z = 5$), oxygen ($Z = 8$), and phosphorus ($Z = 15$).

*2-4. Calculate the energy required for the transition of an electron from the K shell to the L shell in tungsten (see Margin Figure 2-4). Compare the result with the energy necessary for a similar transition in hydrogen. Explain the difference.

*2-5. What is the energy equivalent to the mass of an electron? Because the mass of a particle increases with velocity, assume that the electron is at rest.

*2-6. The energy released during the atomic explosion at Hiroshima was estimated to be equal to that released by 20,000 tons of TNT. Assume that a total energy of 200 MeV is released during fission of a ^{235}U nucleus and that a total energy of 3.8×10^9 J is released during detonation of 1 ton of TNT. Find the number of fissions that occurred in the Hiroshima explosion, and determine the total decrease in mass.

*2-7. A "4-megaton thermonuclear explosion" means that a nuclear explosion releases as much energy as that liberated during detonation of 4 million tons of TNT. Using 3.8×10^9 J/ton as the heat of detonation for TNT, calculate the total energy in joules and in kilocalories released during the nuclear explosion (1 kcal = 4186 J).

2-8. Group the following atoms as isotopes, isotones,and isobars: $^{131}_{54}$Xe, $^{132}_{54}$Xe, $^{130}_{53}$I, $^{133}_{54}$Xe, $^{131}_{53}$I, $^{129}_{52}$Te, $^{132}_{53}$I, $^{130}_{52}$Te, $^{131}_{52}$Te.

SUMMARY

- Atoms are composed of protons and neutrons in the nucleus, which is surrounded by electrons in shell configurations.
- Electron shells:

Shell	Maximum number of Electrons	Binding energy
K ($N = 1$)	2	Highest
L ($N = 2$)	8	Lower than K
M ($N = 3$)	18	Lower than L

- Transitions of electrons from outer shells to inner shells result in the release of characteristic photons and/or auger electrons
- Materials may be classified in terms of their electrical conductivity as
  - Insulators
  - Semiconductors
  - Conductors
  - Superconductors
- The standard form to denote a nuclide is

$$^{A}_{Z}X$$

  where
  A is the mass number, the sum of the number of neutrons and protons
  Z is the number of protons
  X is the chemical symbol (which is uniquely determined by Z)
- The four fundamental fources, in order of strength from strongest to weakest are
  - Nuclear
  - Electrostatic
  - Weak
  - Gravitational
- Nuclear fission and fusion both result in product(s) with higher binding energy per nucleon
- Nuclear nomenclature:
  - Isoto*p*es—same number of *p*rotons
  - Isoto*n*es—same number of *n*eutrons
  - Isob*a*rs—same mass number, *A*
  - Isom*e*rs—same *e*verything *e*xcept *e*nergy

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CHAPTER

3

RADIOACTIVE DECAY

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OBJECTIVES

By studying this chapter, the reader should be able to:

- Understand the relationship between nuclear instability and radioactive decay.
- Describe the different modes of radioactive decay and the conditions in which they occur.
- Draw and interpret decay schemes.
- Write balanced reactions for radioactive decay.
- State and use the fundamental equations of radioactive decay.
- Perform elementary computations for sample activities.
- Comprehend the principles of transient and secular equilibrium.
- Discuss the principles of the artificial production of radionuclides.
- Find information about particular radioactive species.

Radioactivity was discovered in 1896 by Henri Becquerel,[1] who observed the emission of radiation (later shown to be beta particles) from uranium salts. A sentence from his 1896 publication reads "We may then conclude from these experiments that the phosphorescent substance in question emits radiations which penetrate paper opaque to light and reduces the salts of silver." Becquerel experienced a skin burn from carrying a radioactive sample in his vest pocket. This is the first known bioeffect of radiation exposure.

This chapter describes *radioactive decay*, a process whereby unstable nuclei become more stable. All nuclei with atomic numbers greater than 82 are unstable (a solitary exception is $^{209}_{83}$Bi). Many lighter nuclei (i.e., with $Z < 82$) are also unstable. These nuclei undergo radioactive decay (they are said to be "radioactive"). Energy is released during the decay of radioactive nuclei. This energy is termed the *transition energy*.

The transition energy released during radioactive decay is also referred to as the "energy of decay."

NUCLEAR STABILITY AND DECAY

Additional models of the nucleus have been proposed to explain other nuclear properties. For example, the "liquid drop" (also known as the "collective") model was proposed by the Danish physicist Niels Bohr[2] to explain nuclear fission. The model uses the analogy of the nucleus as a drop of liquid.

"Bohr's work on the atom was the highest form of musicality in the sphere of thought". A. Einstein as quoted in Moore, R. *Niels Bohr. The Man, His Science and the World They Changed.* New York, Alfred Knopf, 1966.

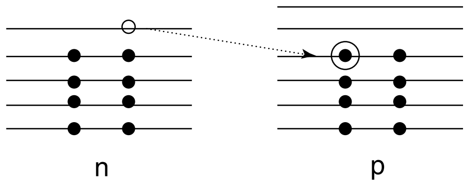
The nucleus of an atom consists of neutrons and protons, referred to collectively as nucleons. In a popular model of the nucleus (the "shell model"), the neutrons and protons reside in specific levels with different binding energies. If a vacancy exists at a lower energy level, a neutron or proton in a higher level may fall to fill the vacancy. This transition releases energy and yields a more stable nucleus. The amount of energy released is related to the difference in binding energy between the higher and lower levels. The binding energy is much greater for neutrons and protons inside the nucleus than for electrons outside the nucleus. Hence, energy released during nuclear transitions is much greater than that released during electron transitions.

If a nucleus gains stability by transition of a neutron between neutron energy levels, or a proton between proton energy levels, the process is termed an *isomeric transition*. In an isomeric transition, the nucleus releases energy without a change in its number of protons (Z) or neutrons (N). The initial and final energy states of the nucleus are said to be *isomers*. A common form of isomeric transition is *gamma decay*, in which the energy is released as a packet of energy (a quantum or photon) termed a gamma (γ) ray. An isomeric transition that competes with gamma decay is *internal conversion*, in which an electron from an extranuclear shell carries the energy out of the atom.

Neutrons can be transformed to protons, and vice versa, by rearrangement of their constituent quarks.

It is also possible for a neutron to fall to a lower energy level reserved for protons, in which case the neutron becomes a proton. It is also possible for a proton to fall to a lower energy level reserved for neutrons, in which case the proton becomes a neutron. In these situations, referred to collectively as *beta* (β) *decay*, the Z and N of the nucleus change, and the nucleus transmutes from one element to another.

In all of the transitions described above, the nucleus loses energy and gains stability. Hence, they are all forms of radioactive decay. In any radioactive process the mass number of the decaying (parent) nucleus equals the sum of the mass numbers of the product (progeny) nucleus and the ejected particle. That is, mass number A is conserved in radioactive decay.



MARGIN FIGURE 3-1
Shell model for the ^{16}N nucleus. A more stable energy state is achieved by an $n \rightarrow p$ transition to form ^{16}O.

ALPHA DECAY

Some heavy nuclei gain stability by a different form of radioactive decay, termed *alpha* (α) *decay*. In this mode of decay, an alpha particle (two protons and two neutrons tightly bound as a nucleus of helium $^4_2$He) is ejected from the unstable nucleus. The alpha particle is a relatively massive, poorly penetrating type of radiation that can be stopped by a sheet of paper. An example of alpha decay is

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

This example depicts the decay of naturally occurring radium into the inert gas radon by emission of an alpha particle. A decay scheme (see below) for alpha decay is depicted in the right margin.

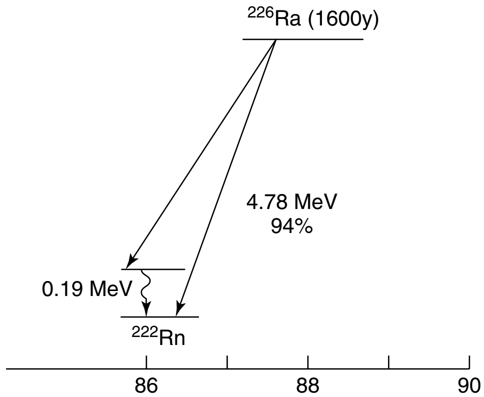
Parent and progeny nuclei were referred to in the past as "mother" and "daughter." The newer and preferred terminology of parent and progeny are used in this text.

Alpha decay was discovered by Marie and Pierre Curie[3] in 1898 in their efforts to isolate radium, and it was first described by Ernest Rutherford[4] in 1899. Alpha particles were identified as helium nuclei by Boltwood and Rutherford in 1911.[5] The Curies shared the 1902 Nobel Prize in Physics with Henri Becquerel.

DECAY SCHEMES

A decay scheme depicts the decay processes specific for a *nuclide* (nuclide is a generic term for any nuclear form). A decay scheme is essentially a depiction of nuclear mass energy on the *y* axis, plotted against the atomic number of the nuclide on the *x* axis. A decay scheme is depicted in Figure 3-1, where the generic nuclide ^A_ZX has four possible routes of radioactive decay:

1. α decay to the progeny nuclide $^{A-4}_{Z-2}$P by emission of a $^4_2$He nucleus;
2. (a) β^+ (positron) decay to the progeny nuclide $_{Z-1}^{\ \ A}$Q by emission of a positive electron from the nucleus;
   (b) electron capture (ec) decay to the progeny nuclide $_{Z-1}^{\ \ A}$Q by absorption of an extranuclear electron into the nucleus;
3. β^-(negatron) decay to the progeny nuclide $_{Z+1}^{\ \ A}$R by emission of a negative electron from the nucleus.



MARGIN FIGURE 3-2
Radioactive decay scheme for α-decay of ^{226}Ra.

The processes denoted as 2(a) and 2(b) are competing pathways to the same progeny nuclide. Any of the pathways can yield a nuclide that undergoes an internal shuffling of nucleons to release additional energy. This process, termed an isomeric transition, is shown as pathway 4. No change in *Z* (or *N* or *A*) occurs during an isomeric transition.

Sometimes the symbol *Q* is added to the right side of the decay reaction to symbolize energy released during the decay process.

After a lifetime of scientific productivity and two Nobel prizes, Marie Curie died in Paris at the age of 67 from aplastic anemia, probably the result of years of exposure to ionizing radiation.

A radionuclide is a radioactive form of a nuclide.

Rutherford, revered as a teacher and research mentor, was known as "Papa" to his many students.

A decay scheme is a useful way to depict and assimilate the decay characteristics of a radioactive nuclide.

Negative and positive electrons emitted during beta decay are created at the moment of decay. They do not exist in the nucleus before decay.

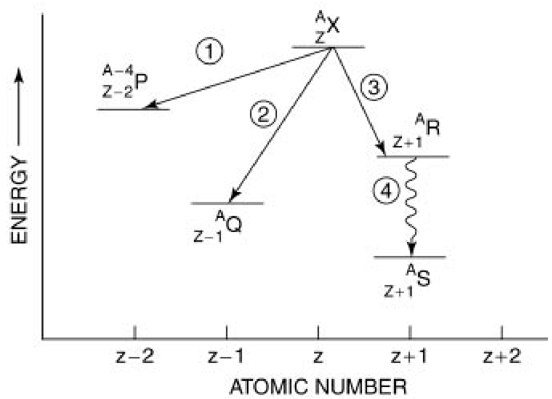


FIGURE 3-1
Radioactive decay scheme.

TABLE 3-1 Nuclear Stability Is Reduced When an Odd Number of Neutrons or Protons Is Present

Number of Protons	*Number of Neutrons*	*Number of Stable Nuclei*
Even	Even	165
Even	Odd	57
Odd	Even	53
Odd	Odd	6

Ernest Rutherford first characterized beta decay in 1899.[4]

The *shell model* of the nucleus was introduced by Maria Goeppert in 1942 to explain the magic numbers.

BETA DECAY

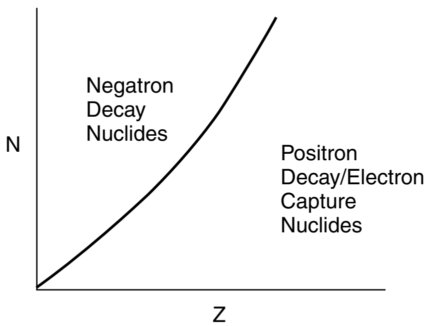
Nuclear Stability

Nuclei tend to be most stable if they contain even numbers of protons and neutrons, and least stable if they contain an odd number of both. This feature is depicted in Table 3-1.

Nuclei are extraordinarily stable if they contain 2, 8, 14, 20, 28, 50, 82, or 126 protons or similar numbers of neutrons. These numbers are termed nuclear *magic numbers* and reflect full occupancy of nuclear shells.

The number of neutrons is about equal to the number of protons in low-Z stable nuclei. As Z increases, the number of neutrons increases more rapidly than the number of protons in stable nuclei, as depicted in Figure 3-2. The shell model of the nucleus accounts for this finding by suggesting that at higher Z, the energy difference is slightly less between neutron levels than between proton levels.

Many nuclei exist that have too many or too few neutrons to reside on or close to the line of stability depicted in Figure 3-2. These nuclei are unstable and undergo radioactive decay. Nuclei above the line of stability (i.e., the n/p ratio is too high for stability) tend to emit negatrons by the process of β^- decay. Nuclei below the line of stability (i.e., the n/p ratio is too low for stability) tend to undergo the competing processes of positron (β^+) decay and electron capture.



MARGIN FIGURE 3-3
Neutron number N and atomic number Z in stable nuclei.

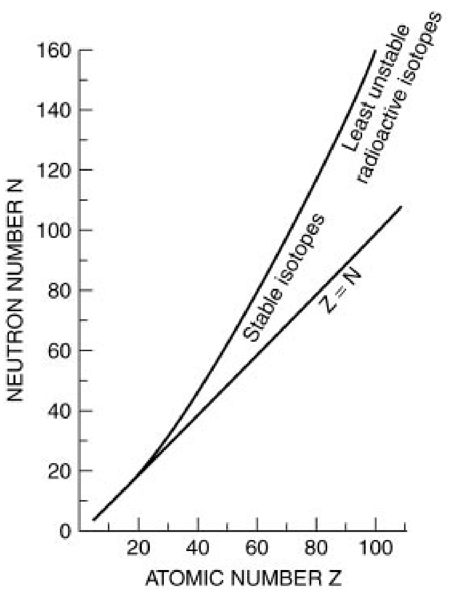


FIGURE 3-2
Number of neutrons (N) for stable (or least unstable) nuclei plotted as a function of the number of protons (Z).

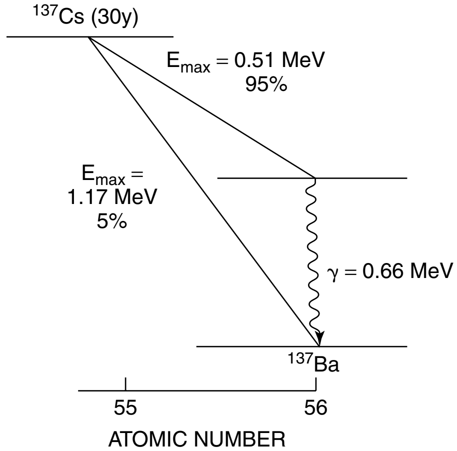
Negatron Decay

In nuclei with an n/p ratio too high for stability, a neutron ${}^1_0n$ may be transformed into a proton ${}^1_0p$:

$$ {}^1_0n \rightarrow {}^1_1p + {}^{\ 0}_{-1}\beta + \tilde{\nu} $$

where ${}^{\ 0}_{-1}\beta$ is a negative electron ejected from the nucleus, and $\tilde{\nu}$ is a massless neutral particle termed an antineutrino (see below). The progeny nucleus has an additional proton and one less neutron than the parent. Therefore, the negatron form of beta decay results in an increase in Z of one, a decrease in N of one, and a constant A. A representative negatron transition is

$$ {}^{137}_{55}\text{Cs} \rightarrow {}^{137}_{56}\text{Ba} + {}^{\ 0}_{-1}\beta + \tilde{\nu} + \text{isomeric transition} $$



MARGIN FIGURE 3-4
Radioactive decay scheme for negatron decay of ^{137}Cs.

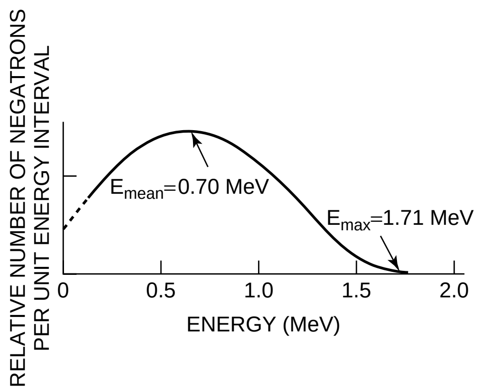
A decay scheme for this transition is shown to the right. A negatron with a maximum energy (E_{max}) of 1.17 MeV is released during 5% of all decays; in the remaining 95%, a negatron with an E_{max} of 0.51 MeV is accompanied by an isomeric transition of 0.66 MeV, where either a γ ray is emitted or an electron is ejected by internal conversion. The transition energy is 1.17 MeV for the decay of ^{137}Cs.

The probability that a nuclide will decay along a particular pathway is known as the branching ratio for the pathway.

Negatron decay pathways are characterized by specific maximum energies; however, most negatrons are ejected with energies lower than these maxima. The average energy of negatrons is about $\frac{1}{3}$ E_{max} along a specific pathway. The energy distribution of negatrons emitted during beta decay of ^{32}P is shown to the right. Spectra of similar shape, but with different values of E_{max} and E_{mean}, exist for the decay pathways of every negatron-emitting radioactive nuclide. In each particular decay, the difference in energy between E_{max} and the specific energy of the negatron is carried away by the antineutrino. That is,

$$ E_{\tilde{\nu}} = E_{max} - E_k $$

where E_{max} is the energy released during the negatron decay process, E_k is the kinetic energy of the negatron, and $E_{\tilde{\nu}}$ is the energy of the antineutrino.



MARGIN FIGURE 3-5
Energy spectrum for negatrons from ^{32}P.

The difference in the energy released during decay, and that possessed by the negatron, threatened the concept of energy conservation for several years. In 1933 Wolfgang Pauli[6] suggested that a second particle was emitted during each decay that accounted for the energy not carried out by the negatron. This particle was named the *neutrino* (Italian for "little neutral particle") by Enrico Fermi.

Positron Decay and Electron Capture

Positron decay results from the nuclear transition

$$ {}^1_1p \rightarrow {}^1_0n + {}^{\ 0}_{+1}\beta + \nu $$

where ${}^{\ 0}_{+1}\beta$ represents a positron ejected from the nucleus during decay, and ν is a neutrino that accompanies the positron. The neutrino and antineutrino are similar, except that they have opposite spin and are said to be antiparticles of each other.

In positron decay, the n/p ratio increases; hence, positron-emitting nuclides tend to be positioned below the n/p stability curve shown in Figure 3-2. Positron decay results in a decrease of one in Z, an increase of one in N, and no change in A. The decay of ${}^{30}_{15}$P is representative of positron decay:

$$ {}^{30}_{15}\text{P} \rightarrow {}^{30}_{14}\text{Si} + {}^{\ 0}_{+1}\beta + \nu $$

The n/p ratio of a nuclide may also be increased by electron capture (ec), in which an electron is captured by the nucleus to yield the transition

$$ {}^1_1p + {}^{\ 0}_{-1}e \rightarrow {}^1_0n + \nu $$

Most electrons are captured from the K electron shell, although occasionally an electron may be captured from the L shell or a shell even farther from the nucleus. During electron capture, a hole is created in an electron shell deep within the atom. This vacancy is filled by an electron cascading from a higher shell, resulting in the release of characteristic radiation or one or more Auger electrons.

Enrico Fermi was a physicist of astounding insight and clarity who directed the first sustained man-made nuclear chain reaction on December 2, 1942 in the squash court of the University of Chicago stadium. He was awarded the 1938 Nobel Prize in Physics.

The antineutrino was detected experimentally by Reines and Cowan[7] in 1953. They used a 10-ton water-filled detector to detect antineutrinos from a nuclear reactor at Savannah River, SC. Reines shared the 1995 Nobel Prize in Physics.

The emission of positrons from radioactive nuclei was discovered in 1934 by Irene Curie[8] (daughter of Marie Curie) and her husband Frederic Joliet. In bombardments of aluminum by α particles, they documented the following transmutation:

$$ {}^{4}_{2}\mathrm{He} + {}^{27}_{13}\mathrm{Al} \rightarrow {}^{30}_{15}\mathrm{P} + {}^{1}_{0}n $$
$$ {}^{30}_{15}\mathrm{P} \rightarrow {}^{30}_{14}\mathrm{Si} + {}^{0}_{+1}\beta + \nu $$

Electron capture of K-shell electrons is known as K-capture; electron capture of L-shell electrons is known as L-capture, and so on.

Many nuclei decay by both electron capture and positron emission, as illustrated in Margin Figure 3-6 for ${}^{22}_{11}\mathrm{Na}$.

$$ {}^{22}_{11}\mathrm{Na} \left\langle \begin{array}{l} + {}^{0}_{-1}e \rightarrow {}^{22}_{10}\mathrm{Ne} + \nu \\ \rightarrow {}^{0}_{+1}\beta + {}^{22}_{10}\mathrm{Ne} + \nu \end{array} \right. $$

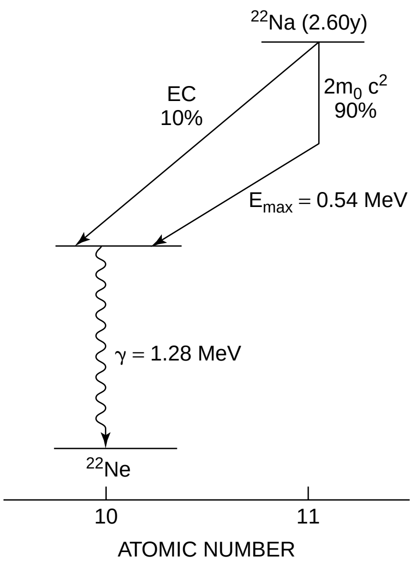
The electron capture branching ratio is the probability of electron capture per decay for a particular nuclide. For ${}^{22}\mathrm{Na}$, the branching ratio is 10% for electron capture, and 90% of the nuclei decay by the process of positron emission. Generally, positron decay prevails over electron capture when both decay modes occur.

A decay scheme for ${}^{22}\mathrm{Na}$ is shown in Figure 3-3. The $2m_0c^2$ listed alongside the vertical portion of the positron decay pathway represents the energy equivalent (1.02 MeV or $2m_0c^2$) of the additional mass of the products of positron decay. This additional mass includes the greater mass of the neutron as compared with the proton, together with the mass of the ejected positron. This amount of energy must be supplied by the transition energy during positron decay. Nuclei that cannot furnish at least 1.02 MeV for the transition do not decay by positron emission. These nuclei increase their n/p ratios solely by electron capture.

A few nuclides may decay by negatron emission, positron emission, or electron capture, as shown in Figure 3-3 for ${}^{74}\mathrm{As}$. This nuclide decays

32% of the time by negatron emission
30% of the time by positron emission
38% of the time by electron capture

All modes of decay result in transformation of the highly unstable "odd–odd" ${}^{74}\mathrm{As}$ nucleus ($Z = 33$, $N = 41$) into a progeny nucleus with even Z and even N.



MARGIN FIGURE 3-6
Radioactive decay scheme for positron decay and electron capture of ${}^{22}\mathrm{Na}$.

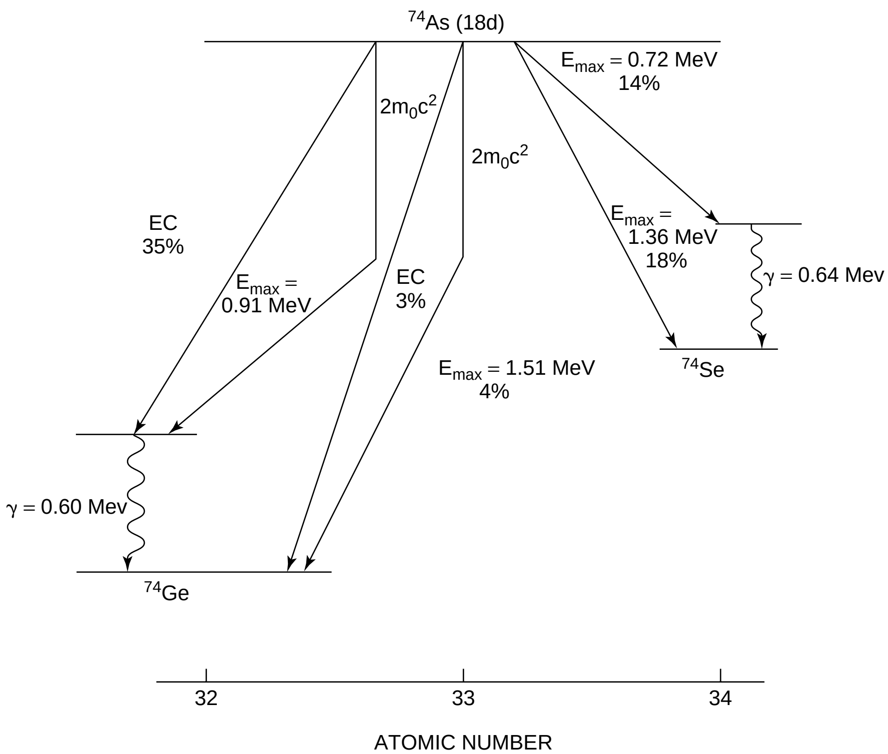


FIGURE 3-3
The decay scheme for ${}^{74}\mathrm{As}$ illustrates competing processes of negatron emission, positron emission, and electron capture.

ISOMERIC TRANSITIONS

As mentioned earlier, radioactive decay often forms a progeny nucleus in an energetic ("excited") state. The nucleus descends from its excited to its most stable ("ground") energy state by one or more isomeric transitions. Often these transitions occur by emission of electromagnetic radiation termed γ rays. γ rays and x rays occupy the same region of the electromagnetic energy spectrum, and they are differentiated only by their origin: x rays result from electron interactions outside the nucleus, whereas γ rays result from nuclear transitions.

No radioactive nuclide decays solely by an isomeric transition. Isomeric transitions are always preceded by either electron capture or emission of an α or $\beta(+ \text{ or } -)$ particle.

Sometimes one or more of the excited states of a progeny nuclide may exist for a finite lifetime. An excited state is termed a *metastable state* if its half-life exceeds 10^{-6} seconds. For example, the decay scheme for ^{99}Mo shown to the right exhibits a metastable energy state, ^{99m}Tc, that has a half-life of 6 hours.

Nuclides emit γ rays with characteristic energies. For example, photons of 142 and 140 keV are emitted by ^{99m}Tc, and photons of 1.17 and 1.33 MeV are released during negatron decay of ^{60}Co. In the latter case, the photons are released during cascade isomeric transitions of progeny ^{60}Ni nuclei from excited states to the ground energy state.

An isomeric transition can also occur by interaction of the nucleus with an electron in one of the electron shells. This process is known as internal conversion (IC). When IC happens, the electron is ejected with kinetic energy E_k equal to the energy E_γ released by the nucleus, reduced by the binding energy E_b of the electron.

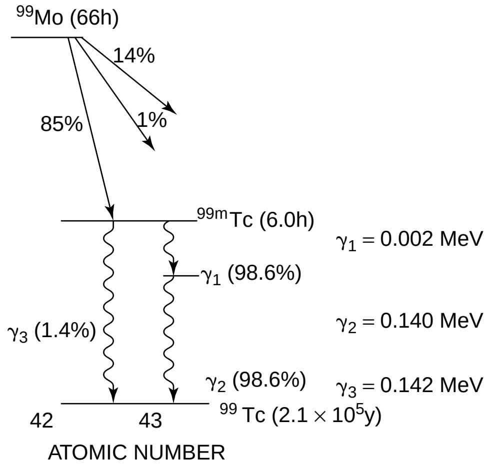
$$E_k = E_\gamma - E_b$$

The ejected electron is accompanied by x rays and Auger electrons as the extranuclear structure of the atom resumes a stable configuration.

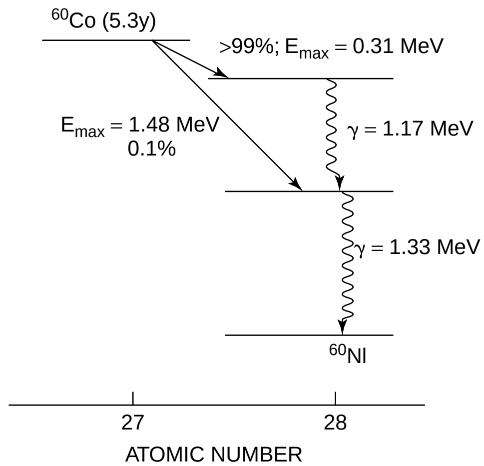
The internal conversion coefficient for an electron shell is the ratio of the number of conversion electrons from the shell compared with the number of γ rays emitted by the nucleus. The probability of internal conversion increases rapidly with increasing Z and with the lifetime of the excited state of the nucleus.

The types of radioactive decay are summarized in Table 3-2.

Gamma rays were discovered by the French physicist Paul Villard in 1900.[9] Rutherford and Andrade confirmed in 1912 that γ rays and x rays are similar types of radiation.



MARGIN FIGURE 3-7
Simplified radioactive decay scheme for ^{99}Mo.



MARGIN FIGURE 3-8
Radioactive decay scheme for ^{60}Co.

MATHEMATICS OF RADIOACTIVE DECAY

Radioactivity can be described mathematically without regard to specific mechanisms of decay. The rate of decay (number of decays per unit time) of a radioactive sample depends on the number N of radioactive atoms in the sample. This concept can be

TABLE 3-2 Summary of Radioactive Decay, $^{A}_{Z}$ (Parent) → $^{A}_{Z}$ (Progeny)

Type of Decay	*A*	*Z*	*N*	*Comments*
Negatron (β^-)	A	$Z+1$	$N-1$	$[E_\beta]_{mean} \cong \frac{[E_\beta]_{max}}{3}$
Positron (β^+)	A	$Z-1$	$N+1$	$[E_\beta]_{mean} \cong \frac{[E_\beta]_{max}}{3}$
Electron capture	A	$Z-1$	$N+1$	Characteristic + auger electrons
Isomeric transition				
Gamma (γ) emission	A	Z	N	Metastable if $T_{1/2} > 10^{-6}$ sec
Internal conversion (IC)	A	Z	N	IC electrons: characteristic + auger electrons
Alpha (α)	$A-4$	$Z-2$	$N-2$	

Equation (3-1) describes the expected decay rate of a radioactive sample. At any moment the actual decay rate may differ somewhat from the expected rate because of statistical fluctuations in the decay rate.

Equation (3-1) depicts a reaction known as a *first-order reaction*.

The decay constant λ is also called the *disintegration constant*.

The decay constant of a nuclide is truly a constant; it is not affected by external influences such as temperature and pressure or by magnetic, electrical, or gravitational fields.

The rutherford (Rf) was once proposed as a unit of activity, where 1 Rf = 10^6 dps. The Rf did not gain acceptance in the scientific community and eventually was abandoned. In so doing, science lost an opportunity to honor one of its pioneers.

The curie was defined in 1910 as the activity of 1 g of radium. Although subsequent measures revealed that 1 g of radium has a decay rate of 3.61 × 10^{10} dps, the definition of the curie was left as 3.7 × 10^{10} dps.

^{201}Tl is an accelerator-produced radioactive nuclide that is used in nuclear medicine to study myocardial perfusion.

stated as

$$\Delta N/\Delta t = -\lambda N \tag{3-1}$$

where $\Delta N/\Delta t$ is the rate of decay, and the constant λ is called the *decay constant*. The minus sign indicates that the number of parent atoms in the sample, and therefore the number decaying per unit time, is decreasing.

By rearranging Eq. (3-1) to

$$\frac{-\Delta N/\Delta t}{N} = \lambda$$

the decay constant can be seen to be the fractional rate of decay of the atoms. The decay constant has units of (time)$^{-1}$, such as sec^{-1} or hr^{-1}. It has a characteristic value for each nuclide. It also reflects the nuclide's degree of instability; a larger decay constant connotes a more unstable nuclide (i.e., one that decays more rapidly). The rate of decay is a measure of a sample's *activity*, defined as

$$\text{Activity} = A = -\Delta N/\Delta t = \lambda N$$

The activity of a sample depends on the number of radioactive atoms in the sample and the decay constant of the atoms. A sample may have a high activity because it contains a few highly unstable (large decay constant) atoms, or because it contains many atoms that are only moderately unstable (small decay constant).

The SI unit of activity is the *becquerel (Bq)*, defined as

$$1 \text{ Bq} = 1 \text{disintegration per second (dps)}$$

Multiples of the becquerel such as the kilobecquerel (kBq = 10^3 Bq), megabecquerel (MBq = 10^6 Bq), and gigabecquerel (GBq = 10^9 Bq) are frequently encountered.

An older, less-preferred unit of activity is the curie (Ci), defined as

$$1 \quad \text{Ci} = 3.7 \times 10^{10} \text{ dps}$$

Useful multiples of the curie are the megacurie (MCi = 10^6 Ci), kilocurie (kCi = 10^3 Ci), millicurie (mCi = 10^{-3} Ci), microcurie (μ Ci = 10^{-6} Ci), nanocurie (nCi = 10^{-9} Ci), and femtocurie (fCi = 10^{-12} Ci).

Example 3-1

a. $^{201}_{81}T$ 1has a decay constant of 9.49 × 10^{-3} hr^{-1}. Find the activity in becquerels of a sample containing 10^{10} atoms. From Eq. (3-1),

$$\begin{aligned} A &= \lambda N \\ &= \frac{(9.49 \times 10^{-3}\,\text{hr}^{-1})(10^{10}\text{ atoms})}{3600\,\text{sec}/\text{hr}} \\ &= 2.64 \times 10^4 \ \frac{\text{atoms}}{\text{sec}} \\ &= 2.64 \times 10^4 \text{ Bq} \end{aligned}$$

b. How many atoms of $^{11}_{6}$C with a decay constant of 2.08 hr^{-1} would be required to obtain the same activity as the sample in Part a?

$$\begin{aligned} N &= \frac{A}{\lambda} \\ &= \frac{2.64 \times 10^4 \frac{\text{atoms}}{\text{sec}} \left(3600 \frac{\text{sec}}{\text{hr}}\right)}{(2.08/\text{hr})} \\ &= 4.57 \times 10^7 \text{ atoms} \end{aligned}$$

More atoms of $^{201}_{81}$T1 than of $^{11}_{6}$C are required to obtain the same activity because of the difference in decay constants.

DECAY EQUATIONS AND HALF-LIFE

Equation (3-1) is a differential equation. It can be solved (see Appendix I) to yield an expression for the number N of parent atoms present in the sample at any time t:

$$N = N_0 e^{-\lambda t} \tag{3-2}$$

where N_0 is the number of atoms present at time $t = 0$, and e is the exponential quantity 2.7183. By multiplying both sides of this equation by λ, the expression can be rewritten as

$$A = A_0 e^{-\lambda t} \tag{3-3}$$

where A is the activity remaining after time t, and A_0 is the original activity.

The number of atoms N^* decaying in time t is $N_0 - N$ (see Figure 3-4), or

$$N^* = N_0(1 - e^{-\lambda t}) \tag{3-4}$$

The probability that any particular atom will not decay during time t is N/N_0, or

$$P\,(\text{no decay}) = e^{-\lambda t} \tag{3-5}$$

and the probability that a particular atom will decay during time t is

$$P\,(\text{decay}) = 1 - e^{-\lambda t} \tag{3-6}$$

For small values of λt, the probability of decay can be approximated as

$$P\,(\text{decay}) \approx \lambda t$$

or expressed per unit time, P (decay per unit time) $\approx \lambda$. That is, when λt is very small, the decay constant approximates the probability of decay per unit time. In most circumstances, however, the approximation is too inexact, and the probability of decay must be computed as $1 - e^{-\lambda t}$. The decay constant should almost always be thought of as a fractional rate of decay rather than as a probability of decay.

The *physical half-life* $T_{1/2}$ of a radioactive nuclide is the time required for decay of half of the atoms in a sample of the nuclide. In Eq. (3-2), $N = \frac{1}{2}N_0$ when $t = T_{1/2}$, with the assumption that $N = N_0$ when $t = 0$. By substitution in Eq. (3-2),

$$\frac{1}{2} = \frac{N}{N_0} = e^{-\lambda T_{1/2}}$$

$$\ln\left(\frac{1}{2}\right) = -\lambda T_{1/2}$$

$$T_{1/2} = \frac{0.693}{\lambda}$$

where 0.693 is the ln 2 (natural logarithm of 2).

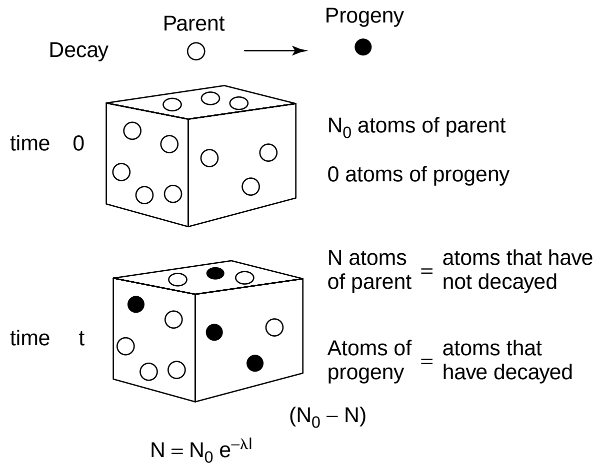
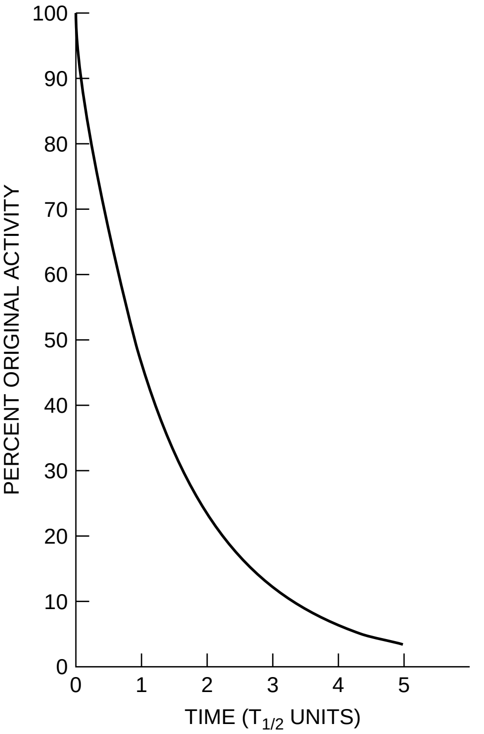


FIGURE 3-4
Mathematics of radioactive decay with the parent decaying to a stable progeny.

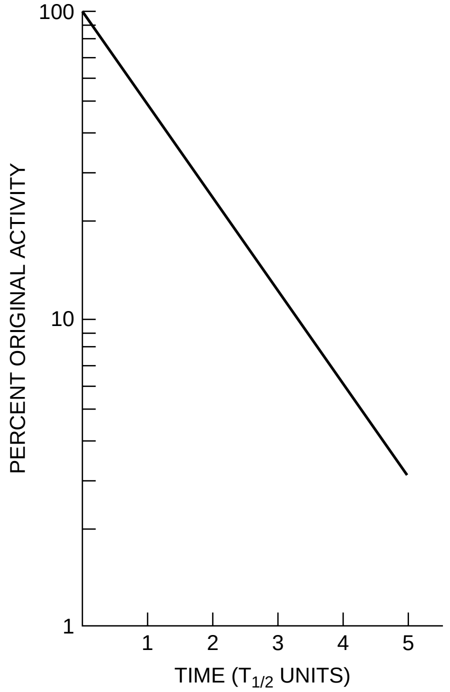
Equation (3-2) reveals that the number N of parent atoms decreases *exponentially* with time.

The *specific activity* of a radioactive sample is the activity per unit mass of sample.

Radioactive decay must always be described in terms of the probability of decay; whether any particular radioactive nucleus will decay within a specific time period is never certain.



MARGIN FIGURE 3-9
The percent of original activity of a radioactive sample is expressed as a function of time in units of half-life.



MARGIN FIGURE 3-10
Semilogarithmic graph of the data in Margin Figure 3-9.

^{113}In was used at one time, but no longer, as a blood pool imaging agent.

The mean (average) life is the average expected lifetime for atoms of a radioactive sample. The *mean life* τ is related to the decay constant λ by

$$\tau = \frac{1}{\lambda}$$

Because $T_{1/2} = \frac{(0.693)}{\lambda}$ the mean life may be calculated from the expression

$$\tau = 1.44 T_{1/2}$$

The percent of original activity of a radioactive sample is pictured in Margin Figure 3-9 as a function of time expressed in units of physical half-life. By replotting these data semilogarithmically (activity on a logarithmic axis, time on a linear axis), a straight line can be obtained, as shown in Margin Figure 3-10.

Example 3-2

The physical half-life is 1.7 hours for ^{113m}In.

a. A sample of ^{113m}In has a mass of 2 μg. How many ^{113m}In atoms are present in the sample?

$$\text{Number of atoms } N = \frac{\text{(No. of grams)(No of atoms/gram} - \text{atomic mass)}}{\text{No. of grams/gram} - \text{atomic mass}}$$

$$= \frac{(2 \times 10^{-6}\ \text{g})(6.02 \times 10^{23}\ \text{atoms/gram} - \text{atomic mass})}{113\ \text{grams/gram} - \text{atomic mass}}$$

$$= 1.07 \times 10^{16}\ \text{atoms}$$

b. How many ^{113m}In atoms remain after 4 hours have elapsed?

$$\text{Number of atoms remaining} = N = N_0 e^{-\lambda t}$$

Since $\lambda = 0.693/T_{1/2}$

$$N = N_0 e^{-(0.693/T_{1/2})t}$$
$$= (1.07 \times 10^{16}\ \text{atoms})\, e^{-(0693/1.7\ \text{hr})\ 4.0\ \text{hr}}$$
$$= (1.07 \times 10^{16}\ \text{atoms})(0.196)$$
$$= 2.10 \times 10^{15}\ \text{atoms remaining}$$

c. What is the activity of the sample when $t = 4.0$ hours?

$$A = \lambda N = \frac{0.693}{T_{1/2}} N$$
$$= \frac{0.693(2.1 \times 10^{15}\ \text{atoms})}{1.7\ \text{hr}\,(3600\ \text{sec/hr})}$$
$$= 2.4 \times 10^{11}\ \text{dps}$$

In units of activity

$$A = 2.4 \times 10^{11}\ \text{Bq}$$

Because of the short physical half-life (large decay constant) of ^{113m}In, a very small mass of this nuclide possesses high activity.

d. Specific activity is defined as the activity per unit mass of a radioactive sample. What is the specific activity of the ^{113m}In sample after 4 hours?

$$\text{Specific activity} = \frac{2.4 \times 10^{11}\ \text{Bq}}{2 \times 10^{-6}\ \text{g}} = 1.2 \times 10^{17}\ \frac{\text{Bq}}{\text{g}}$$

e. Enough ^{113m}In must be obtained at 4 p.m. Thursday to provide 300 kBq at 1 p.m. Friday. How much ^{113m}In should be obtained?

$$A = A_0 e^{-\lambda t}$$

$$A_0 = Ae^{\lambda t} = Ae^{0.693t/T_{1/2}}$$

$$= (300\ \text{kBq})e^{(0.693)(21\ \text{hr})/1.7\ \text{hr}}$$

$$A = 1.57 \times 10^9\ \text{Bq} = 1.57\ \text{GBq} = \text{activity that should be obtained at 4 p.m. Thursday}$$

Example 3-3 may be solved graphically by plotting the activity of each sample as a function of time. The time when activities are equal is indicated by the intersection of the curves for the nuclides.

Equation (3-7) is a Bateman equation. More complex Bateman equations describe progeny activities for sequential phases of multiple radioactive nuclides in transient equilibrium.

Example 3-3

The half-life of ^{99m}Tc is 6.0 hours. How much time t must elapse before a 100-GBq sample of ^{113m}In and a 20-GBq sample of ^{99m}Tc possess equal activities?

$$A^{99m\text{Tc}} = A^{113m} \text{ at time } t$$

$$A_{0(^{99m}\text{Tc})} e^{-(0.693)t/(T_{1/2})^{99m}\text{Tc}} = A_{0(^{113m}\text{In})} e^{-(0.693)t/(T_{1/2})113m\text{In}}$$

$$(20\text{GBq})e^{-(0.693)t/6.0\ \text{hr}} = (100\ \text{GBq})e^{-(0.693)t/1.7\ \text{hr}}$$

$$\frac{100\ \text{GBq}}{20\ \text{GBq}} = e^{(0.408-0.115)t}$$

$$5 = e^{(0.408-0.115)t}$$

$$t = 5.5\ \text{hr before activities are equal}$$

In the ^{132}I–^{132}Te example, transient equilibrium occurs

- at only one instant in time
- when ^{132}Te reaches its maximum activity
- when the activity of ^{132}Te in the sample is neither increasing nor decreasing
- when the activities of ^{132}I and ^{132}Te are equal.

^{99m}Tc is the most widely used radioactive nuclide in nuclear medicine. It is produced in a ^{99}Mo–^{99m}Tc radioactive generator, often called a molybdenum "cow." Eluting ^{99m}Tc from the generator is referred to as "milking the cow." The ^{99}Mo–^{99m}Tc generator was developed by Powell Richards in 1957.

TRANSIENT EQUILIBRIUM

Progeny atoms produced during radioactive decay are sometimes also radioactive. For example, ^{226}Ra decays to ^{222}Rn ($T_{1/2} = 3.83$ days); and ^{222}Rn decays by α-emission to ^{218}Po, an unstable nuclide that in turn decays with a half-life of about 3 minutes. If the half-life of the parent is significantly longer than the half-life of the progeny (i.e., by a factor of 10 or more), then a condition of *transient equilibrium* may be established. After the time required to attain transient equilibrium has elapsed, the activity of the progeny decreases with an apparent half-life equal to the physical half-life of the parent. The apparent half-life reflects the simultaneous production and decay of the progeny atoms. If no progeny atoms are present initially when $t = 0$, the number N_2 of progeny atoms present at any later time is

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_0 (e^{-\lambda_1 t} - e^{-\lambda_2 t}) \tag{3-7}$$

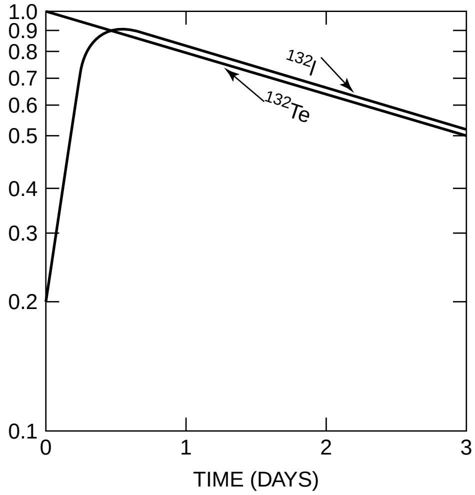
In Equation (3-7), N_0 is the number of parent atoms present when $t = 0$, λ_1 is the decay constant of the parent, and λ_2 is the decay constant of the progeny. If $(N_2)_0$ progeny atoms are present when $t = 0$, then Eq. (3-7) may be rewritten as

$$N_2 = (N_2)_0 e^{-\lambda_2 t} + \frac{\lambda_1}{\lambda_2 - \lambda_1} N_0 (e^{-\lambda_1 t} - e^{-\lambda_2 t})$$

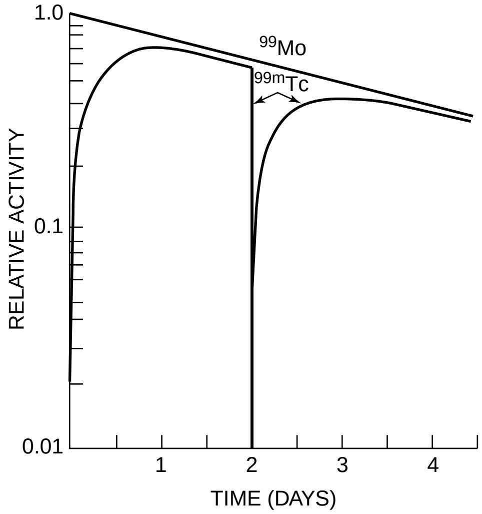
Transient equilibrium for the transition

$$^{132}\text{Te}\left(T_{1/2} = 78\ \text{hr}\right) \rightarrow {}^{132}\text{I}\left(T_{1/2} = 2.3\ \text{hr}\right)$$

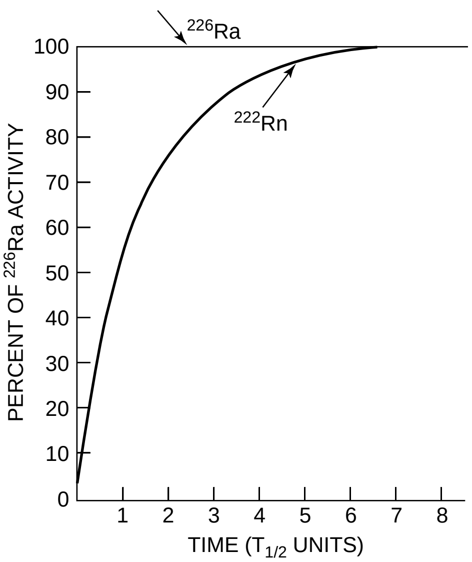
is illustrated in the Margin Figure 3-11. The activity of ^{132}I is greatest at the moment of transient equilibrium when parent (^{132}Te) and progeny (^{132}I) activities are equal. At all later times, the progeny activity exceeds the activity of the parent (A_1).



MARGIN FIGURE 3-11
Activities of parent ^{132}Te and progeny ^{132}I as a function of time illustrate the condition of transient equilibrium that may be achieved when the half-life of the parent is not much greater than the half-life of the progeny.



MARGIN FIGURE 3-12
Activities of parent (^{99}Mo) and progeny (^{99m}Tc) in a nuclide generator. About 14% of the parent nuclei decay without the formation of ^{99m}Tc.



MARGIN FIGURE 3-13
Growth of activity and secular equilibrium of ^{222}Rn produced by decay of ^{226}Ra.

At these times the progeny (A_2) can be shown to equal

$$\frac{A_1}{A_2} = \frac{\lambda_2 - \lambda_1}{\lambda_2}$$

Transient equilibrium is the principle underlying production of short-lived isotopes (e.g., ^{99m}Tc, ^{113m}In, and ^{68}Ga) in generators for radioactive nuclides used in nuclear medicine. For example, the activities of ^{99m}Tc ($T_{1/2} = 6.0$ hours and ^{99}Mo ($T_{1/2} =$ 66 hours) are plotted in Margin Figure 3-12 as a function of time. The ^{99m}Tc activity remains less than that for ^{99}Mo because about 14% of the ^{99}Mo nuclei decay promptly to ^{99}Tc without passing through the isomeric ^{99m}Tc state. The abrupt decrease in activity at 48 hours reflects the removal of ^{99m}Tc from the generator.

Secular Equilibrium

Secular equilibrium may occur when the half-life of the parent is much greater than the half-life of the progeny (e.g., by a factor of 10^4 or more). Since $\lambda_1 \ll \lambda_2$ if $T_{1/2\ \text{parent}} \gg T_{1/2\ \text{progeny}}$, Eq. (3-7) may be simplified by assuming that $\lambda_2 - \lambda_1 = \lambda_2$ and that $e^{-\lambda_1 t} = 1$.

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_0\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right) \cong \frac{\lambda_1}{\lambda_2} N_0\left(1 - e^{-\lambda_2 t}\right)$$

After several half-lives of the daughter have elapsed, $e^{-\lambda_2 t} \cong 0$ and

$$N_2 = \frac{\lambda_1}{\lambda_2}(N_0)$$

$$\lambda_2 N_2 = \lambda_1 N_0 \tag{3-8}$$

$$\frac{0.693}{\left(T_{1/2}\right)_2} N_2 = \frac{0.693}{\left(T_{1/2}\right)_1} N_0$$

$$\frac{N_2}{\left(T_{1/2}\right)_2} = \frac{N_0}{\left(T_{1/2}\right)_1} \tag{3-9}$$

The terms $\lambda_1 N_0$ and $\lambda_2 N_2$ in Eq. (3-8) describe the activities of parent and progeny, respectively. These activities are equal after secular equilibrium has been achieved. Illustrated in Margin Figure 3-13 are the growth of activity and the activity at equilibrium for ^{222}Rn produced in a closed environment by decay of ^{226}Ra. Units for the x axis are multiples of the physical half-life of ^{222}Rn. The growth curve for ^{222}Rn approaches the decay curve for ^{226}Ra asymptotically. Several half-lives of ^{222}Rn must elapse before the activity of the progeny equals 99% of the parent activity.

If the parent half-life is less than that for the progeny ($T_{1/2}$ parent $<$ $T_{1/2}$ progeny, or $\lambda_1 > \lambda_2$), then a constant relationship is not achieved between the activities of the parent and progeny. Instead, the activity of the progeny increases initially, reaches a maximum, and then decreases with a half-life intermediate between the parent and progeny half-lives.

Radioactive nuclides in secular equilibrium are often used in radiation oncology. For example, energetic β-particles from ^{90}Y in secular equilibrium with ^{90}Sr are used to treat intraocular lesions. The activity of the ^{90}Sr–^{90}Y ophthalmic irradiator decays with the physical half-life of ^{90}Sr (28 years), whereas a source of ^{90}Y alone decays with the 64-hour half-life of ^{90}Y. Radium needles and capsules used widely in the past for superficial and intracavitary radiation treatments contain many decay products in secular equilibrium with long-lived ^{226}Ra.

Natural Radioactivity and Decay Series

Almost every radioactive nuclide found in nature is a member of one of three radioactive decay series. Each series consists of sequential transformations that begin with a long-lived parent and end with a stable nuclide. In a closed environment, products

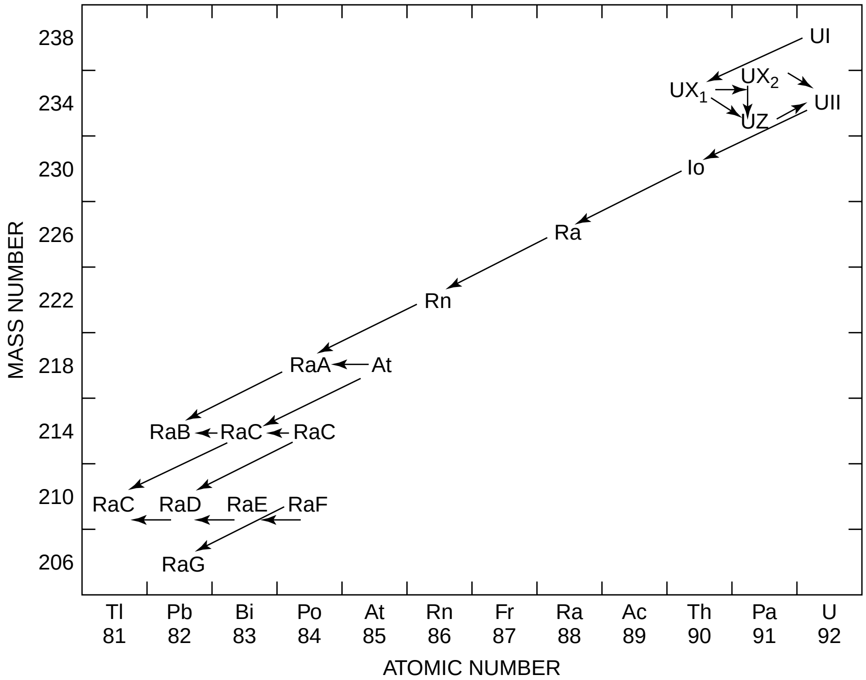


FIGURE 3-5
Uranium ($4n + 2$) radioactive decay series.

of intermediate transformations within each series are in secular equilibrium with the long-lived parent. These products decay with an apparent half-life equal to the half-life of the parent. All radioactive nuclides found in nature decay by emitting either α-particles or negatrons. Consequently, each transformation in a radioactive decay series changes the mass number of the nucleus either by 4 (α-decay) or by 0 (negatron decay).

The uranium series begins with ^{238}U ($T_{1/2} = 4.5 \times 10^9$ years) and ends with stable ^{206}Pb. The mass number 238 of the parent nuclide is divisible by 4 with a remainder of 2. All members of the uranium series, including stable ^{206}Pb, also possess mass numbers divisible by 4 with remainder of 2. Consequently, the uranium decay series sometimes is referred to as the "$4n + 2$ series," where n represents an integer between 51 and 59 (Figure 3-5). The nuclide ^{226}Ra and its decay products are members of the uranium decay series. A sample of ^{226}Ra decays with a half-life of 1600 years. However, 4.5×10^9 years are required for the earth's supply of ^{226}Ra to decrease to half because ^{226}Ra in nature is in secular equilibrium with its parent ^{238}U.

Other radioactive series are the actinium or "$4n + 3$ series" ($^{235}U \rightarrow\rightarrow$ ^{207}Pb) and the thorium or "$4n$ series" ($^{232}Th \rightarrow\rightarrow$ ^{208}Pb). Members of the hypothetical neptunium or "$4n + 1$ series" ($^{241}Am \rightarrow\rightarrow$ ^{209}Bi) are not found in nature because there is no long-lived parent for this series.

Fourteen naturally occurring radioactive nuclides are not members of a decay series. The 14 nuclides, all with relatively long half-lives, are ^{3}H, ^{14}C,^{40}K, ^{50}V, ^{87}Rb, ^{115}In, ^{130}Te, ^{138}La, ^{142}Ce, ^{144}Nd, ^{147}Sm, ^{176}Lu, and ^{187}Re, and ^{192}Pt.

The inert gas ^{222}Rn produced by decay of naturally occurring ^{226}Ra is also radioactive, decaying with a half-life of 3.83 days. This radioactive gas first called "radium emanation" was characterized initially by Rutherford. Seepage of ^{222}Rn into homes built in areas with significant ^{226}Ra concentrations in the soil is an ongoing concern to homeowners and the Environmental Protection Agency.

ARTIFICIAL PRODUCTION OF RADIONUCLIDES

Nuclides may be produced artificially that have properties desirable for medicine, research, or other purposes. Nuclides with excess neutrons (which will therefore emit negatrons) are created by bombarding nuclei with neutrons from a nuclear reactor, The first high-energy particle accelerator was the cyclotron developed by Ernest Lawrence in 1931. In 1938, Ernest and his physician brother John used artificially produced ^{32}P to treat their mother, who was afflicted with leukemia.

Nuclear transmutation by particle bombardment was first observed by Rutherford[10] in 1919 in his studies of α-particles traversing an air-filled chamber. The observed transmutation was

$$^{4}_{2}He + ^{14}_{7}N \rightarrow ^{17}_{8}O + ^{1}_{1}H$$

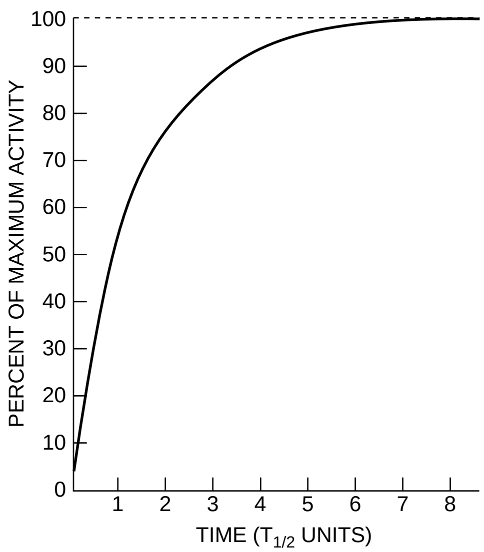
where He represents α-particles and H depicts protons detected during the experiment. This reaction can be written more concisely as

$$^{14}_{7}N(\alpha, p)^{17}_{8}O$$

where $^{14}_{7}N$ represents the bombarded nucleus, $^{17}_{8}O$ the product nucleus, and (α, p) the incident and ejected particles, respectively.

Through their discovery of artificial radioactivity, Irene Curie (the daughter of Marie Curie) and Federic Joliot paved the way to use of radioactive tracers in biomedical research and clinical medicine.

The cross-sectional area presented to a neutron by a target nucleus is usually described in units of barns, with 1 barn equaling 10^{-24} cm^2.



MARGIN FIGURE 3-14
Growth in activity for a sample of ^{59}Co bombarded by slow neutrons in a nuclear reactor.

for example,

$$^{13}_{6}C + ^{1}_{0}n \rightarrow ^{14}_{6}C$$
$$^{31}_{15}P + ^{1}_{0}n \rightarrow ^{32}_{15}P$$

Other isotopes produced by this method include

$$^{3}_{1}H,\ ^{35}_{16}S,\ ^{51}_{24}Cr,\ ^{60}_{27}Co,\ ^{99}_{42}Mo,\ ^{133}_{154}Xe,\ ^{198}_{79}Au$$

Nuclides with excess protons may be produced by bombarding nuclei with protons, α-particles, deuterons, or other charged particles from high-energy particle accelerators. These nuclides then decay by positron decay or electron capture with the emission of positrons, photons, and so on. Representative transitrons include:

$$^{68}_{30}Zn + ^{1}_{1}p \rightarrow ^{67}_{31}Ga + 2n$$
$$^{11}_{5}B + ^{1}_{1}p \rightarrow ^{11}_{6}C + n$$

Useful radioisotopes produced by particle accelerators include

$$^{13}_{7}N,\ ^{15}_{8}O,\ ^{18}_{9}F$$

Other nuclides are obtained as by-products of fission in nuclear reactors; for example,

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{A_1}_{Z_1}x + ^{A_2}_{Z_2}y + \text{neutrons}$$

where a range of isotopes X and Y are produced, including

$$^{90}_{38}Sr,\ ^{99}_{42}Mo,\ ^{131}_{53}I,\ ^{137}_{55}Cs$$

Finally, some radionuclides are produced as decay products of isotopes produced by the above methods. For example, $^{201}_{82}Pb$ is produced by bombardment of $^{203}_{81}T1$ by protons.

$$^{203}_{81}Tl + p \rightarrow ^{201}_{82}Pb + 3n$$

$^{201}_{82}Pb(T_{1/2} = 9.4$ hr) decays by electron capture to the useful radionuclide $^{201}_{81}T1$.

MATHEMATICS OF NUCLIDE PRODUCTION BY NEUTRON BOMBARDMENT

The production of artificially radioactive nuclides by neutron bombardment in the core of a nuclear reactor may be described mathematically. The activity A of a sample bombarded for a time t, assuming no radioactivity when $t = 0$, may be written as

$$A = \varphi N\sigma(1 - e)^{-(0.693/T_{1/2})^t} \qquad \textbf{(3-10)}$$

where

- φ is the neutron flux in neutrons per square centimeter-second
- N is the number of target nuclei in the sample
- σ is the absorption cross section of target nuclei in square centimeters
- $T_{1/2}$ is the half-life of the product nuclide, and
- A is the activity of the sample in becquerels

When the bombardment time t is much greater than $T_{1/2}$, Eq. (3-10) reduces to

$$A_{max} = \varphi N\sigma \qquad \textbf{(3-11)}$$

where A_{max} represents the maximum activity in becquerels. Margin Figure 3-14 illustrates the growth of radioactivity in a sample of ^{59}Co bombarded by slow neutrons. This transmutation is written $^{59}_{27}Co\,(n, \gamma)\,^{60}_{27}Co$. The half-life of ^{60}Co is 5.3 years.

Ru $99^{5/+}$ 12.72 σ 11 98.90594	Ru 100 12.62 σ 5.8 99.90422	Ru $101^{5/+}$ 17.07 σ 5.2 100.90558
Tc 98 1.5×10^6y β^-.3 γ.75, .56 σ(3+?) E1.7	Tc $99^{9/+}$ 6.0h IT .002, .142, γ.140 2.1×10^5y β^- .29 E.29 σ.22	Tc 100^{1+} 17s β^-3.37, 2.24,$^{++}$ γ.54, .59,$^{++}$ E3.37
Mo $97^{5/+}$ 9.46 σ 2 96.90602	Mo 98 23.78 σ .51 97.90541	Mo 99 67h β^-123, .45,$^{++}$ γ(.14,++), .74, .041–.95 E1.37

MARGIN FIGURE 3-15
Section from a chart on the nuclides.

Example 3-4

A 20-g sample of ^{59}Co is positioned in the core of a reactor with an average neutron flux of 10^{14} neutrons/cm^2 sec. The absorption cross section of ^{59}Co is 36 barns.

a. What is the activity of the sample after 6.0 years?

$$N = \frac{20\text{ g}\,(6.02\times10^{23}\text{ atoms/gram} - \text{atomic mass}}{59\text{ grams/gram} - \text{atomic mass}} = 2.04\times10^{23}\text{ atoms}$$

$$A = \varphi N\sigma\left(1 - e^{-(0.693)t/T_{1/2}}\right)$$

$$A = (10^{14}\text{neutrons/cm}^2\text{- sec})(2.04\times10^{23}\text{ atoms})(36\times10^{-24}\text{ cm}^2)$$
$$\times\left[1 - e^{-(0.693)6.0\text{ yrs}/5.3\text{ yr}}\right]$$
$$= 4.00\times10^{14}\text{ Bq}$$

b. What is the maximum activity for the sample?

$$A_{\max} = \varphi N\sigma$$
$$A = (10^{14}\text{ neutrons/cm}^2\text{-sec})(2.04\times10^{23}\text{ atoms})(36\times10^{-24}\text{ cm}^2)$$
$$= 7.33\times10^{14}\text{ Bq}$$

c. When does the sample activity reach 90% of its maximum activity?

$$(7.33\times10^{14}\text{ Bq})(0.9) = 6.60\times10^{14}\text{ Bq}$$
$$6.60\times10^{14}\text{ Bq} = 7.33\times10^{14}\text{ Bq}\left[1 - e^{-(0.693)t/5.3\text{ yr}}\right]$$
$$6.60\times10^{14}\text{ Ci} = 7.33\times10^{14} - 7.33\times10^{14}\,e^{-0.131t}$$
$$7.33\times10^{14}\,e^{-0.131} = 0.73\times10^{14}$$
$$e^{-0.131t} = 0.10$$
$$0.131t = 2.30$$
$$t = 17.6\text{ years to reach 90\% }A_{\max}$$

INFORMATION ABOUT RADIOACTIVE NUCLIDES

Decay schemes for many radioactive nuclei are listed in the *Radiological Health Handbook,*[11] *Table of Isotopes,*[12] and Medical Internal Radiation Dose Committee pamphlets published by the Society of Nuclear Medicine.[13]

Charts of the nuclides (available from General Electric Company, Schenectady, New York, and from Mallinckrodt Chemical Works, St. Louis) contain useful data concerning the decay of radioactive nuclides. A section of a chart is shown to the right. In this chart, isobars are positioned along 45-degree diagonals, isotones along vertical lines, and isotopes along horizonal lines. The energy of radiations emitted by various nuclei is expressed in units of MeV.

The term "isotope" was suggested to the physical chemist Frederick Soddy by the physician and novelist Margaret Todd over dinner in the Glasgow home of Soddy's father-in-law.

PROBLEMS

*3-1. The half-life of ^{132}I is 2.3 hours. What interval of time is required for 3.7×10^9 Bq of ^{132}I to decay to 9.25×10^8 Bq? What interval of time is required for decay of 7/8 of the ^{132}I atoms? What interval of time is required for 3.7×10^9 Bq of ^{132}I to decay to 9.25×10^8 Bq if the ^{132}I is in transient equilibrium with ^{132}Te($T_{1/2} = 78$ hours)?

*3-2. What is the mass in grams of 3.7×10^9 Bq of pure ^{32}P ($T_{1/2} =$ 14.3 days)? How many ^{32}P atoms constitute 3.7×10^9 Bq? What is the mass in grams of 3.7×10^9 Bq of $Na_3{}^{32}PO_4$ if all the phosphorus in the compound is radioactive?

3-3. Some ^{210}Bi nuclei decay by α-emission, whereas others decay by negatron emission. Write the equation for each mode of decay, and identify the daughter nuclide.

*For problems marked with an asterisk, answers are provided on page 491.

*3-4. If a radioactive nuclide decays for an interval of time equal to its average life, what fraction of the original activity remains?

3-5. From the decay scheme for ^{131}I, determine the branching ratio for the mode of negatron emission that results in the release of γ rays of 364 keV. From the decay scheme for ^{126}I, determine the branching ratios for the emission of negatrons and positrons with different maximum energies. Determine the branching ratios of ^{126}I for positron emission and electron capture.

*3-6. What are the frequency and wavelength of a 100-keV photon?

3-7. ^{126}I nuclei decay may be negatron emission, positron emission, or electron capture. Write the equation for each mode of decay, and identify the daughter nuclide.

*3-8. How much time is required before a 3.7×10^8 Bq sample of ^{99m}Tc ($T_{1/2} = 6.0$ hours) and a 9.25×10^8 Bq sample of ^{113m}In ($T_{1/2} = 1.7$ hours) possess equal activities.

3-9. From a chart of the nuclides determine the following:
a. Whether ^{202}Hg is stable or unstable
b. Whether ^{193}Hg decays by negatron or positron emission
c. The nuclide that decays to ^{198}H by positron emission
d. The nuclide that decays to ^{198}Hg by negatron emission
e. The half-life of ^{203}Hg
f. The percent abundance of ^{198}Hg in naturally occurring mercury
g. The atomic mass of ^{204}Hg

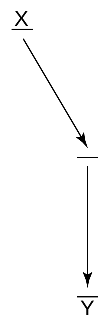
*3-10. How many atoms and grams of ^{90}Y are in secular equilibrium with 1.85×10^9 Bq of ^{90}Sr?

*3-11. How many becquerels of ^{24}Na should be ordered so that the sample activity will be 3.7×10^8 Bq when it arrives 24 hours later?

*3-12. Fifty grams of gold (^{197}Au) are subjected to a neutron flux of 10^{13} neutrons/cm^2-sec in the core of a nuclear reactor. How much time is required for the activity to reach 3.7×10^4 GBq? What is the maximum activity for the sample? What is the sample activity after 20 minutes? The cross section of ^{197}Au is 99 barns.

*3-13. The only stable isotope of arsenic is ^{75}As. What modes of radioactive decay would be expected for ^{74}As and ^{76}As?

*3-14. For a nuclide X with the decay scheme



how many γ rays are emitted per 100 disintegrations of X if the coefficient for internal conversion is 0.25?

SUMMARY

- Radioactive decay is the consequence of nuclear instability:
  - Negatron decay occurs in nuclei with a high n/p ratio.
  - Positron decay and electron capture occur in nuclei with a low n/p ratio.
  - Alpha decay occurs with heavy unstable nuclei.
  - Isomeric transitions occur between different energy states of nuclei and result in the emission of γ rays and conversion electrons.
- The activity A of a sample is

$$A = A_0 e^{-\lambda t}$$

  where λ is the decay constant (fractional rate of decay).
- The half-life $T_{1/2}$ is

$$T_{1/2} = 0.693/\lambda$$

- The common unit of activity is the becquerel (Bq), with 1 Bq = 1 disintegration/second.
- Transient equilibrium may exist when the progeny nuclide decays with a $T_{1/2} <$ $T_{1/2}$ parent.
- Secular equilibrium may exist when the progeny nuclide decays with a $T_{1/2} \ll$ $T_{1/2}$ parent
- Most radioactive nuclides found in Nature are members of naturally occurring decay series.

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CHAPTER

4

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OBJECTIVES

After completing this chapter, the reader should be able to:

- Compare directly ionizing and indirectly ionizing radiation and give examples of each.
- Define and state the relationships among linear energy transfer, specific ionization, the *W*-quantity, range, and energy.
- List the interactions that occur for electrons.
- Give examples of elastic and inelastic scattering by nuclei.
- Describe the interactions of neutrons.
- Write an equation for attenuation of photons in a given thickness of a material.
- List the interactions that occur for photons.
- Define atomic, electronic, mass, and linear attenuation coefficients.
- Discuss and compare energy absorption and energy transfer.
- List at least five types of electromagnetic radiation.

Directly and Indirectly Ionizing Radiation

Directly Ionizing Particles (Charged Particles)	Indirectly Ionizing Particles (Uncharged Particles)
Alpha (helium nuclei)	Photons
Any nuclei	Neutrons
Beta (electrons)	
Protons	

The term *interaction* may be used to describe the crash of two automobiles (an example in the macroscopic world) or the collision of an x ray with an atom (an example in the submicroscopic world). This chapter describes radiation interactions on a submicroscopic scale. Interactions in both macroscopic and microscopic scales follow fundamental principles of physics such as (a) the conservation of energy and (b) the conservation of momentum.

CHARACTERISTICS OF INTERACTIONS

In a radiation interaction, the radiation and the material with which it interacts may be considered as a single system. When the system is compared before and after the interaction, certain quantities will be found to be *invariant*. Invariant quantities are exactly the same before and after the interaction. Invariant quantities are said to be *conserved* in the interaction. One quantity that is always conserved in an interaction is the total energy of the system, with the understanding that mass is a form of energy. Other quantities that are conserved include momentum and electric charge.

Some quantities are not always conserved during an interaction. For example, the number of particles may not be conserved because particles may be fragmented, fused, "created" (energy converted to mass), or "destroyed" (mass converted to energy) during an interaction. Interactions may be classified as either *elastic* or *inelastic*. An interaction is elastic if the sum of the *kinetic energies* of the interacting entities is conserved during the interaction. If some energy is used to free an electron or nucleon from a bound state, kinetic energy is not conserved and the interaction is inelastic. Total energy is conserved in all interactions, but kinetic energy is conserved only in interactions designated as elastic.

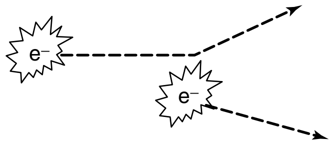
DIRECTLY IONIZING RADIATION

When an electron is ejected from an atom, the atom is left in an *ionized* state. Hydrogen is the element with the smallest atomic number and requires the least energy (binding energy of 13.6 eV) to eject its K-shell electron. Radiation of energy less than 13.6 eV is termed *nonionizing radiation* because it cannot eject this most easily removed electron. Radiation with energy above 13.6 eV is referred to as *ionizing radiation*. If electrons are not ejected from atoms but merely raised to higher energy levels (outer shells), the process is termed *excitation*, and the atom is said to be "excited." Charged particles such as electrons, protons, and atomic nuclei are directly ionizing radiations because they can eject electrons from atoms through charged-particle interactions. Neutrons and photons (x and γ rays) can set charged particles into motion, but they do not produce significant ionization directly because they are uncharged. These radiations are said to be indirectly ionizing.

Energy transferred to an electron in excess of its binding energy appears as kinetic energy of the ejected electron. An ejected electron and the residual positive ion constitute an *ion pair*, abbreviated IP. An average energy of 33.85 eV, termed the W-quantity or W, is expended by charged particles per ion pair produced in air.[1] The average energy required to remove an electron from nitrogen or oxygen (the most common atoms in air) is much less than 33.85 eV. The W-quantity includes not only the electron's binding energy but also the average kinetic energy of the ejected electron and the average energy lost as incident particles excite atoms, interact with nuclei, and increase the rate of vibration of nearby molecules. On the average, 2.2 atoms are excited per ion pair produced in air.

The specific ionization (SI) is the number of primary and secondary ion pairs produced per unit length of path of the incident radiation. The specific ionization of α-particles in air varies from about 3–7 × 10^6 ion pairs per meter, and the specific ionization of protons and deuterons is slightly less. The linear energy transfer (LET) is the average loss in energy per unit length of path of the incident radiation. The LET is the product of the specific ionization and the W-quantity:

$$\text{LET} = (\text{SI})(W) \tag{4-1}$$



MARGIN FIGURE 4-1
One electron is incident upon a target electron. As the incident electron nears the target electron, the electrostatic fields of the two negatively charged electrons interact, with the result that the stationary electron is set in motion by the repulsive force of the incident electron, and the direction of the incident electron is changed. This classical picture assumes the existence of electrostatic fields for the two electrons that exist simultaneously everywhere in space.

Classical Electrodynamics versus Quantum Electrodynamics (QED)
The classical diagram of particle interactions shows a picture of the interaction in space. The Feynman diagram of quantum electrodynamics (QED) shows the development of the interaction in time. The classical diagram of the interaction between two electrons assumes electrostatic fields for the electrons that exist everywhere in space simultaneously. The Feynman QED diagram of the interaction shows the brief exchange of a virtual photon between the two electrons.

In radiologic physics, we are primarily interested in determining which particles survive an interaction, where they go, and where they deposit their energy. For these purposes, the classical diagrams are adequate.

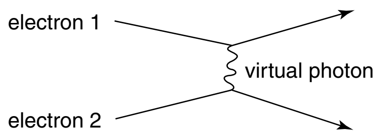
Example 4-1

Assuming that the average specific ionization is 4 × 10^6 ion pairs per meter (IP/m), calculate the average LET of α-particles in air.

$$\begin{aligned}\text{LET} &= (\text{SI})(W)\\ &= 4\times 10^6\ \frac{\text{IP}}{\text{m}} \times 33.85\ \frac{\text{eV}}{\text{IP}}\\ &= 1.35\times 10^5\ \frac{\text{keV}}{\text{m}}\end{aligned}$$

The range of ionizing particles in a particular medium is the straight-line distance traversed by the particles before they are completely stopped. For heavy particles with energy E that tend to follow straight-line paths, the range in a particular medium may be estimated from the average LET:

$$\text{Range} = \frac{E}{\text{LET}} \tag{4-2}$$



MARGIN FIGURE 4-2
The Feynman diagram of quantum electrodynamics (QED) showing the exchange of a virtual photon between two electrons. When electron 1 emits a virtual photon in a downward direction, conservation of momentum requires the electron to recoil upward. When electron 2 absorbs the virtual photon, it gains momentum and, therefore, must recoil downward. In the QED picture, the electrostatic force between two electrons is explained in terms of exchange of a virtual photon that can travel a specific distance in a specific time.

Example 4-2

Calculate the range in air for 4-MeV α-particles with an average LET equal to that computed in Example 4-1.

$$\begin{aligned}\text{Range} &= \frac{E}{\text{LET}}\\ &= \frac{4\text{MeV}(10^3\text{keV/ MeV})}{1.35\times 10^5\text{keV/m}}\\ &= \text{approximately } 0.03\text{ m} = 3\text{ cm in air}\end{aligned}$$

INTERACTIONS OF ELECTRONS

Interactions of negative and positive electrons may be divided into three categories:

1. Interactions with electrons
2. Elastic interactions with nuclei
3. Inelastic interactions with nuclei

Scattering by Electrons

Negative and positive electrons traversing an absorbing medium transfer energy to electrons of the medium. Impinging electrons lose energy and are deflected at some angle with respect to their original direction. An electron receiving energy may be raised to a shell farther from the nucleus or may be ejected from the atom. The kinetic energy E_k of an ejected electron equals the energy E received minus the binding energy E_B of the electron:

$$E_k = E - E_B \tag{4-3}$$

If the binding energy is negligible compared with the energy received, then the interaction may be considered an elastic collision between "free" particles. If the binding energy must be considered, then the interaction is inelastic.

Incident negatrons and positrons are scattered by electrons with a probability that increases with the atomic number of the absorber and decreases rapidly with increasing kinetic energy of the incident particles. Low-energy negatrons and positrons interact frequently with electrons of an absorber; the frequency of interaction diminishes rapidly as the kinetic energy of the incident particles increases.[2]

Ion pairs are produced by negatrons and positrons during both elastic and inelastic interactions. The specific ionization (ion pairs per meter [IP/m]) in air at STP (standard temperature = 0°C, standard pressure = 760 mm Hg) may be estimated with Eq. (4-4) for negatrons and positrons with kinetic energies between 0 and 10 MeV.

$$\text{SI} = \frac{4500}{(v/c)^2} \tag{4-4}$$

In Eq. (4-4), v represents the velocity of an incident negatron or positron and c represents the speed of light *in vacuo* (3×10^8 m/sec).

Of the many interesting individuals who helped shape our modern view of quantum mechanics, Richard Feynman is remembered as one of the most colorful. Recognized as a math and physics prodigy in college, he came to the attention of the physics community for his work on the Manhattan Project in Los Alamos, New Mexico. In addition to his many contributions to atomic physics and computer science during the project, he was known for playing pranks on military security guards by picking locks on file cabinets, opening safes, and leaving suspicious notes. His late-night bongo sessions in the desert of New Mexico outside the lab were also upsetting to security personnel. After the war, he spent most of his career at Cal Tech, where his lectures were legendary, standing-room-only sessions attended by faculty as well as students.

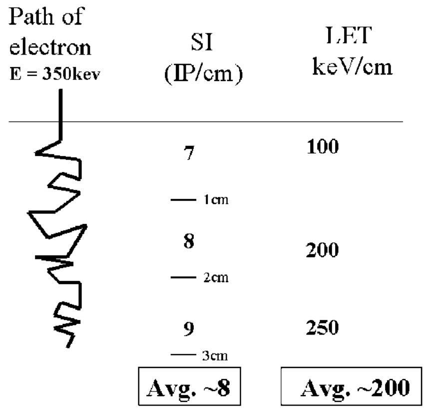
Example 4-3

Calculate the specific ionization (SI) and linear energy transfer (LET) of 0.1 MeV electrons in air ($v/c = 0.548$). The LET may be computed from the SI or Eq. (4-1) by using an average W-quantity for electrons of 33.85 eV/IP:

$$\begin{aligned} \text{SI} &= \frac{4500}{(v/c)^2} \\ &= \frac{4500}{(0.548)^2} \\ &= 15{,}000 \text{ IP/m} \end{aligned}$$

$$\begin{aligned} \text{LET} &= (\text{SI})\,(W) \\ &= (15{,}000 \text{ IP/m})\,(33.85 \text{ eV/IP})\,(10^{-3} \text{ keV/eV}) \\ &= 508 \text{ keV/m} \end{aligned}$$

After expending its kinetic energy, a positron combines with an electron in the absorbing medium. The particles annihilate each other, and their mass appears as electromagnetic radiation, usually two 0.51-MeV photons moving in opposite directions. These photons are termed *annihilation radiation*, and the interaction is referred to as *pair annihilation*.



MARGIN FIGURE 4-3
An electron having energy $E = 350$ keV interacts in a tissue-like material. Its actual path is torturous, changing direction a number of times, as the electron interacts with atoms of the material via excitations and ionizations. As interactions reduce the energy of the electron through excitation and ionization, the elctron's energy is transferred to the material. The interactions that take place along the path of the particle may be summarized as specific ionization (SI, ion pairs/cm) or as linear energy transfer (LET, keV/cm) along the straight line continuation of the particle's trajectory beyond its point of entry.

Elastic Scattering by Nuclei

Electrons are deflected with reduced energy during elastic interactions with nuclei of an absorbing medium. The probability of elastic interactions with nuclei varies with Z^2 of the absorber and approximately with $1/E_k^2$, where E_k represents the kinetic energy of the incident electrons. The probability for elastic scattering by nuclei is slightly less for positrons than for negatrons with the same kinetic energy. Backscattering of negatrons and positrons in radioactive samples is primarily due to elastic scattering by nuclei.

Probabilities for elastic scattering of electrons by electrons and nuclei of an absorbing medium are about equal if the medium is hydrogen ($Z = 1$). In absorbers with higher atomic number, elastic scattering by nuclei occurs more frequently than electron scattering by electrons because the nuclear scattering cross section varies with Z^2, whereas the cross section for scattering by electrons varies with Z.

Inelastic Scattering by Nuclei

A negative or positive electron passing near a nucleus may be deflected with reduced velocity. The interaction is inelastic if energy is released as electromagnetic radiation during the encounter. The radiated energy is known as bremsstrahlung (braking radiation). A bremsstrahlung photon may possess any energy up to the entire kinetic energy of the incident particle. For low-energy electrons, bremsstrahlung photons are radiated predominantly at right angles to the motion of the particles. The angle narrows as the kinetic energy of the electrons increases (see Margin Figure 4-6).

The probability of bremsstrahlung production varies with Z^2 of the absorbing medium. A typical bremsstrahlung spectrum is illustrated in Margin Figure 4-4. The relative shape of the spectrum is independent of the atomic number of the absorber.

The ratio of radiation energy loss (the result of inelastic interactions with nuclei) to the energy lost by excitation and ionization (the result of interactions with electrons) is approximately

$$\frac{\text{Radiation energy loss}}{\text{Ionization energy loss}} = \frac{E_k Z}{820} \tag{4-5}$$

where E_k represents the kinetic energy of the incident electrons in MeV and Z is the atomic number of the absorbing medium. For example, excitation-ionization and bremsstrahlung contribute about equally to the energy lost by 10-MeV electrons traversing lead ($Z = 82$). The ratio of energy lost by the production of bremsstrahlung to that lost by ionization and excitation of atoms is important to the design of x-ray tubes in the diagnostic energy range (below 0.1 MeV) where the ratio is much smaller and therefore x ray production is less efficient.

The speed of light in a vacuum (3×10^8 m/sec) is the greatest velocity known to be possible. Particles cannot exceed a velocity of 3×10^8 m/sec under any circumstances. However, light travels through many materials at speeds slower than its speed in a vacuum. In a material, it is possible for the velocity of a particle to exceed the speed of light in the material. When this occurs, visible light known as Cerenkov radiation is emitted. Cerenkov radiation is the cause of the blue glow that emanates from the core of a "swimming pool"-type nuclear reactor. Only a small fraction of the kinetic energy of high-energy electrons is lost through production of Cerenkov radiation.

A cross section is an expression of probability that an interaction will occur between particles. The bigger the cross section, the higher the probability an interaction will occur. This is the same principle as the observation that "the bigger the target, the easier it is to hit." The unit in which atomic cross sections are expressed is the barn, where one barn equals 10^{-28} m^2. Its origin is in the American colloquialism, "as big as a barn" or "such a bad shot, he couldn't hit the broad side of a barn." The unit name was first used by American scientists during the Manhattan Project, the project in which the atomic bomb was developed during World War II. In 1950, the Joint Commission on Standards, Units, and Constants of Radioactivity recommended international acceptance because of its common usage in the United States (Evans, R. D. *The Atomic Nucleus*. Malabar, FL, Krieger Publishing, 1955, p. 9.)